

Implementation of Parallel Linear Static Solver

1 Introduction

This document catalogues every GMSH element type encountered by a FEM solver dependent on P1, P2 or P3 line, triangular, quadrilateral, tetrahedral, or hexahedral elements. For each type we list:

- the GMSH numerical type ID,
- the local node ordering as written in that line (which reader code interprets directly), and
- a TikZ figure of the canonical reference element with every local index drawn at its lattice position.

2 The `$Elements` block in MSH 2.2

A Gmsh ASCII MSH 2.2 file declares its mesh elements in a block of the form:

```
$Elements
<number-of-elements>
<element-id> <element-type> <number-of-tags> <tag-1> ... <tag-N> <node-1>
    ... <node-K>
...
$EndElements
```

- `element-id` is a element identifier (usually unused by the reader program).
- `element-type` is the integer that selects the shape and order (Table 1).
- `number-of-tags` is typically 2: the first tag is the *physical group* ID (the reader program uses this to classify volume vs. Robin / Neumann / Dirichlet), the second is the *elementary entity* ID (usually unused by the reader program).
- `node-1` ... `node-K` are 1-based global node IDs in the order gmsh prescribes for that element type. K is the total node count (3 for a P1 triangle, 10 for a P3 triangle, etc.).

The reader program's only job in this block is to recognize `element-type`, read exactly K integers, and remap them (where needed) into its preferred in-memory ordering.

3 Summary of element types used

Table 1 lists every element type, together with where it is parsed. Type 15 (1-node point) is recognized in every file but most of the times discarded.

Table 1: GMSH element types.

ID	Shape	Order	Nodes	Used by
15	Point	—	1	usually ignored
1	Line	P1	2	boundary edges in 2D
8	Line	P2	3	boundary edges in 2D P2 triangular or quadrilateral elements
26	Line	P3	4	boundary edges in 2D P3 triangular or quadrilateral elements
2	Triangle	P1	3	domain in P1 triangle; faces in P1 tetrahedral
9	Triangle	P2	6	domain in P2 triangle; faces in P2 tetrahedral
21	Triangle	P3	10	domain in P3 triangle; faces in P3 tetrahedral
3	Quadrilateral	Q1	4	domain in P1 quadrilateral; faces in P1 hexahedral
10	Quadrilateral	Q2	9	domain in P2 quadrilateral; faces in P2 hexahedral
36	Quadrilateral	Q3	16	domain in P3 quadrilateral; faces in P3 hexahedral
4	Tetrahedron	P1	4	domain in P1 tetrahedral
11	Tetrahedron	P2	10	domain in P2 tetrahedral
29	Tetrahedron	P3	20	domain in P3 tetrahedral
5	Hexahedron	Q1	8	domain in P1 hexahedral
12	Hexahedron	Q2	27	domain in P2 hexahedral
92	Hexahedron	Q3	64	domain in P3 hexahedral

4 Line elements

Lines appear as boundary segments in 2D meshes (Robin / Neumann / Dirichlet edges) and as 1D edges of higher-dimensional elements that gmsh writes out even when the reader program ignores them.

4.1 P1 Line Element (type 1, 2 nodes)

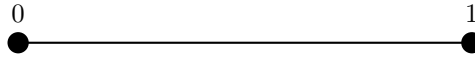


Table 2: Coordinates of nodes on a standard P1 line element.

Node Number	Coordinate
0	-1
1	1

4.2 P2 Line Element (type 8, 3 nodes)

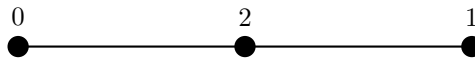


Table 3: Coordinates of nodes on a standard P2 line element.

Node Number	Coordinate
0	-1
1	1
2	0

4.3 P3 Line Element (type 26, 4 nodes)



Table 4: Coordinates of nodes on a standard P3 line element.

Node Number	Coordinate
0	-1
1	1
2	$-\frac{1}{3}$
3	$\frac{1}{3}$

5 Triangle elements

The reference triangle used here has vertices $V_0 = (0,0)$, $V_1 = (1,0)$, $V_2 = (0,1)$, traversed counter-clockwise (gmsh convention).

5.1 P1 Triangle Element (type 2, 3 nodes)

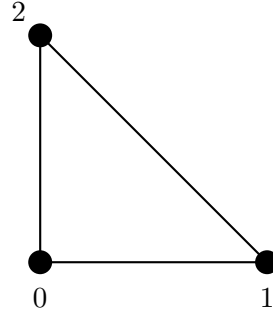


Table 5: Coordinates of nodes on a standard P1 triangle element.

Node Number	Coordinate
0	$(0,0)$
1	$(1,0)$
2	$(0,1)$

5.2 P2 Triangle Element (type 9, 6 nodes)

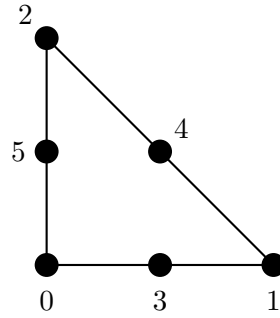


Table 6: Coordinates of nodes on a standard P2 triangle element.

Node Number	Coordinate
0	$(0,0)$
1	$(1,0)$
2	$(0,1)$
3	$(\frac{1}{2}, 0)$
4	$(\frac{1}{2}, \frac{1}{2})$
5	$(0, \frac{1}{2})$

5.3 P3 Triangle Element (type 21, 10 nodes)

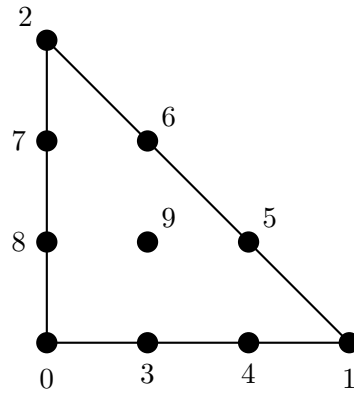


Table 7: Coordinates of nodes on a standard P3 triangle element.

Node Number	Coordinate
0	$(0, 0)$
1	$(1, 0)$
2	$(0, 1)$
3	$(\frac{1}{3}, 0)$
4	$(\frac{2}{3}, 0)$
5	$(\frac{2}{3}, \frac{1}{3})$
6	$(\frac{1}{3}, \frac{2}{3})$
7	$(0, \frac{2}{3})$
8	$(0, \frac{1}{3})$
9	$(\frac{1}{3}, \frac{1}{3})$

6 Quadrilateral elements

The reference quadrilateral has corners $V_0 = (0, 0)$, $V_1 = (1, 0)$, $V_2 = (1, 1)$, $V_3 = (0, 1)$, traversed counter-clockwise.

6.1 Q1 Quadrilateral Element (type 3, 4 nodes)

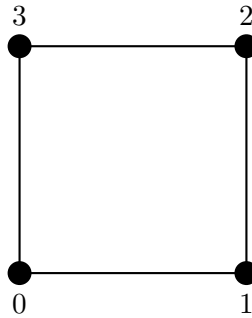


Table 8: Coordinates of nodes on a standard Q1 quadrilateral element.

Node Number	Coordinate
0	(0, 0)
1	(1, 0)
2	(1, 1)
3	(0, 1)

6.2 Q2 Quadrilateral Element (type 10, 9 nodes)

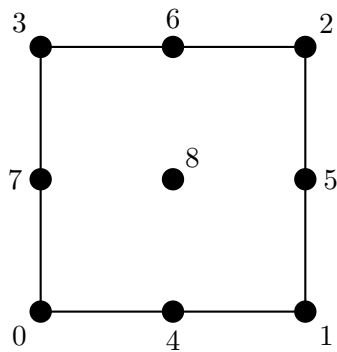


Table 9: Coordinates of nodes on a standard Q2 quadrilateral element.

Node Number	Coordinate
0	$(0, 0)$
1	$(1, 0)$
2	$(1, 1)$
3	$(0, 1)$
4	$(\frac{1}{2}, 0)$
5	$(1, \frac{1}{2})$
6	$(\frac{1}{2}, 1)$
7	$(0, \frac{1}{2})$
8	$(\frac{1}{2}, \frac{1}{2})$

6.3 Q3 Quadrilateral Element (type 36, 16 nodes)

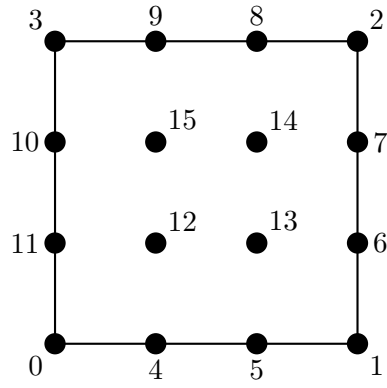


Table 10: Coordinates of nodes on a standard Q3 quadrilateral element.

Node Number	Coordinate
0	$(0, 0)$
1	$(1, 0)$
2	$(1, 1)$
3	$(0, 1)$
4	$(\frac{1}{3}, 0)$
5	$(\frac{2}{3}, 0)$
6	$(1, \frac{1}{3})$
7	$(1, \frac{2}{3})$
8	$(\frac{2}{3}, 1)$
9	$(\frac{1}{3}, 1)$
10	$(0, \frac{2}{3})$
11	$(0, \frac{1}{3})$
12	$(\frac{1}{3}, \frac{1}{3})$
13	$(\frac{2}{3}, \frac{1}{3})$
14	$(\frac{2}{3}, \frac{2}{3})$
15	$(\frac{1}{3}, \frac{2}{3})$

7 Tetrahedral elements

7.1 P1 Tetrahedron Element (type 4, 4 nodes)

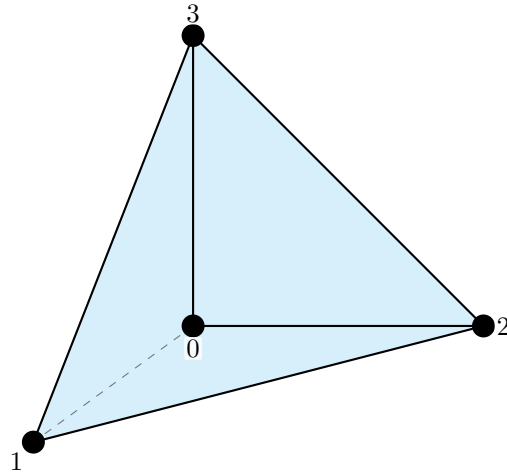


Table 11: Coordinates of nodes on a standard P1 tetrahedron element.

Node Number	Coordinate
0	(0, 0, 0)
1	(1, 0, 0)
2	(0, 1, 0)
3	(0, 0, 1)

7.2 P2 Tetrahedron Element (type 11, 10 nodes)

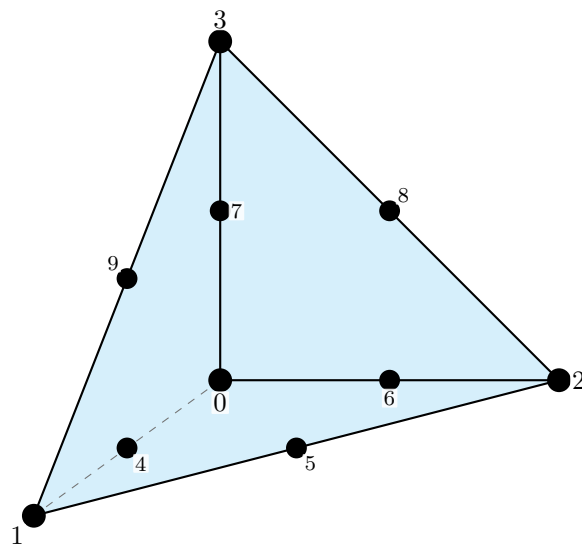


Table 12: Coordinates of nodes on a standard P2 tetrahedron element.

Node Number	Coordinate
0	$(0, 0, 0)$
1	$(1, 0, 0)$
2	$(0, 1, 0)$
3	$(0, 0, 1)$
4	$(\frac{1}{2}, 0, 0)$
5	$(\frac{1}{2}, \frac{1}{2}, 0)$
6	$(0, \frac{1}{2}, 0)$
7	$(0, 0, \frac{1}{2})$
8	$(0, \frac{1}{2}, \frac{1}{2})$
9	$(\frac{1}{2}, 0, \frac{1}{2})$

7.3 P3 Tetrahedron Element (type 29, 20 nodes)

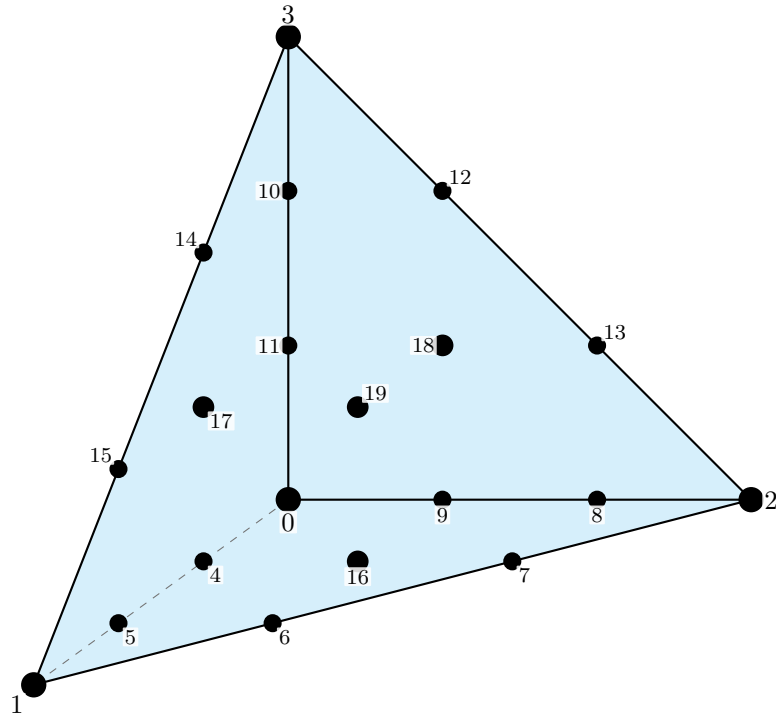


Table 13: Coordinates of nodes on a standard P3 tetrahedron element.

Node Number	Coordinate
0	$(0, 0, 0)$
1	$(1, 0, 0)$
2	$(0, 1, 0)$
3	$(0, 0, 1)$
4	$(\frac{1}{3}, 0, 0)$
5	$(\frac{2}{3}, 0, 0)$
6	$(\frac{2}{3}, \frac{1}{3}, 0)$
7	$(\frac{1}{3}, \frac{2}{3}, 0)$
8	$(0, \frac{2}{3}, 0)$
9	$(0, \frac{1}{3}, 0)$
10	$(0, 0, \frac{2}{3})$
11	$(0, 0, \frac{1}{3})$
12	$(0, \frac{1}{3}, \frac{2}{3})$
13	$(0, \frac{2}{3}, \frac{1}{3})$
14	$(\frac{1}{3}, 0, \frac{2}{3})$
15	$(\frac{2}{3}, 0, \frac{1}{3})$
16	$(\frac{1}{3}, \frac{1}{3}, 0)$
17	$(\frac{1}{3}, 0, \frac{1}{3})$
18	$(0, \frac{1}{3}, \frac{1}{3})$
19	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

8 Hexahedral elements

8.1 Q1 Hexahedron Element (type 5, 8 nodes)

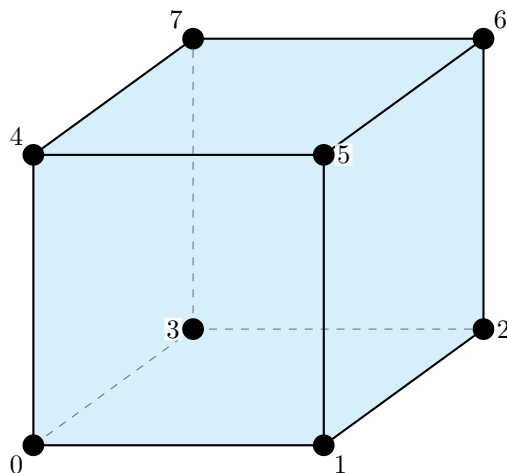


Table 14: Coordinates of nodes on a standard Q1 hexahedron element.

Node Number	Coordinate
0	(0, 0, 0)
1	(1, 0, 0)
2	(1, 1, 0)
3	(0, 1, 0)
4	(0, 0, 1)
5	(1, 0, 1)
6	(1, 1, 1)
7	(0, 1, 1)

8.2 Q2 Hexahedron Element (type 12, 27 nodes)

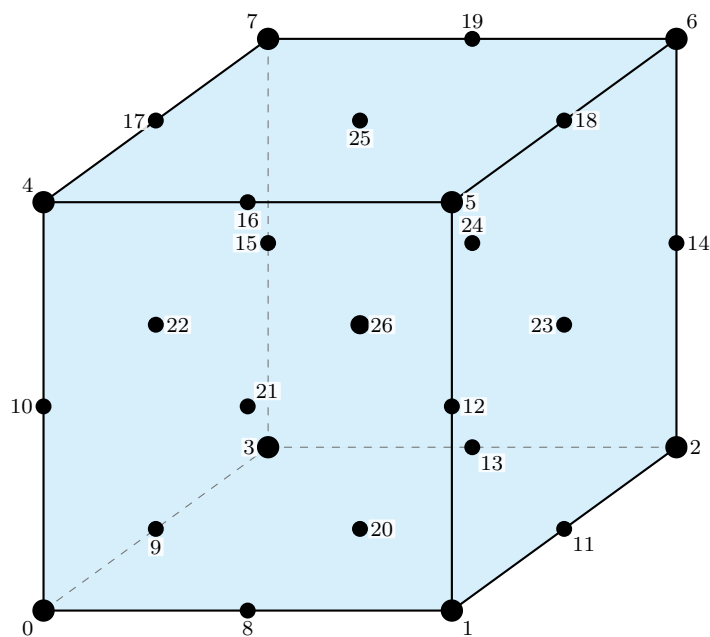


Table 15: Coordinates of nodes on a standard Q2 hexahedron element.

Node Number	Coordinate
0	$(0, 0, 0)$
1	$(1, 0, 0)$
2	$(1, 1, 0)$
3	$(0, 1, 0)$
4	$(0, 0, 1)$
5	$(1, 0, 1)$
6	$(1, 1, 1)$
7	$(0, 1, 1)$
8	$(\frac{1}{2}, 0, 0)$
9	$(0, \frac{1}{2}, 0)$
10	$(0, 0, \frac{1}{2})$
11	$(1, \frac{1}{2}, 0)$
12	$(1, 0, \frac{1}{2})$
13	$(\frac{1}{2}, 1, 0)$
14	$(1, 1, \frac{1}{2})$
15	$(0, 1, \frac{1}{2})$
16	$(\frac{1}{2}, 0, 1)$
17	$(0, \frac{1}{2}, 1)$
18	$(1, \frac{1}{2}, 1)$
19	$(\frac{1}{2}, 1, 1)$
20	$(\frac{1}{2}, \frac{1}{2}, 0)$
21	$(\frac{1}{2}, 0, \frac{1}{2})$
22	$(0, \frac{1}{2}, \frac{1}{2})$
23	$(1, \frac{1}{2}, \frac{1}{2})$
24	$(\frac{1}{2}, 1, \frac{1}{2})$
25	$(\frac{1}{2}, \frac{1}{2}, 1)$
26	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

8.3 Q3 Hexahedron Element (type 92, 64 nodes)

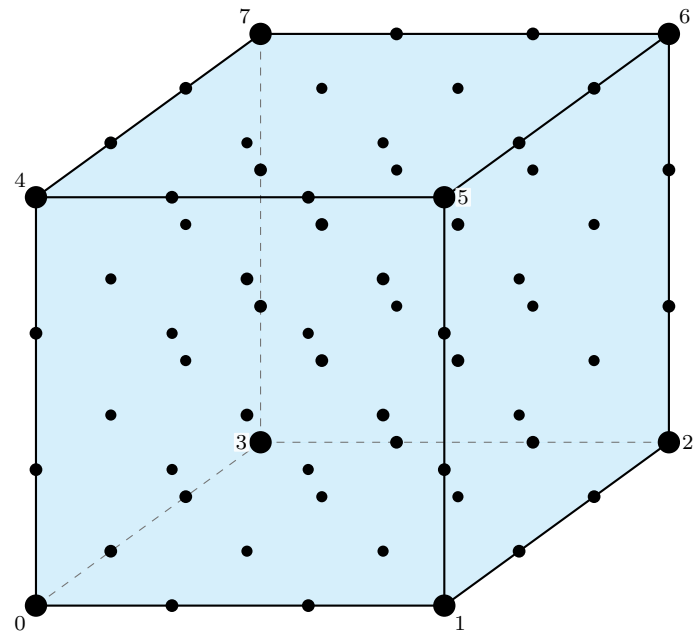


Table 16: Coordinates of nodes on a standard Q3 hexahedron element (corners and edge nodes).

Node Number	Coordinate
0	$(0, 0, 0)$
1	$(1, 0, 0)$
2	$(1, 1, 0)$
3	$(0, 1, 0)$
4	$(0, 0, 1)$
5	$(1, 0, 1)$
6	$(1, 1, 1)$
7	$(0, 1, 1)$
8	$(\frac{1}{3}, 0, 0)$
9	$(\frac{2}{3}, 0, 0)$
10	$(0, \frac{1}{3}, 0)$
11	$(0, \frac{2}{3}, 0)$
12	$(0, 0, \frac{1}{3})$
13	$(0, 0, \frac{2}{3})$
14	$(1, \frac{1}{3}, 0)$
15	$(1, \frac{2}{3}, 0)$
16	$(1, 0, \frac{1}{3})$
17	$(1, 0, \frac{2}{3})$
18	$(\frac{2}{3}, 1, 0)$
19	$(\frac{1}{3}, 1, 0)$
20	$(1, 1, \frac{1}{3})$
21	$(1, 1, \frac{2}{3})$
22	$(0, 1, \frac{1}{3})$
23	$(0, 1, \frac{2}{3})$
24	$(\frac{1}{3}, 0, 1)$
25	$(\frac{2}{3}, 0, 1)$
26	$(0, \frac{1}{3}, 1)$
27	$(0, \frac{2}{3}, 1)$
28	$(1, \frac{1}{3}, 1)$
29	$(1, \frac{2}{3}, 1)$
30	$(\frac{2}{3}, 1, 1)$
31	$(\frac{1}{3}, 1, 1)$

Table 17: Coordinates of nodes on a standard Q3 hexahedron element (face and interior nodes).

Node Number	Coordinate
32	$(\frac{1}{3}, \frac{1}{3}, 0)$
33	$(\frac{2}{3}, \frac{1}{3}, 0)$
34	$(\frac{2}{3}, \frac{2}{3}, 0)$
35	$(\frac{1}{3}, \frac{2}{3}, 0)$
36	$(\frac{1}{3}, 0, \frac{1}{3})$
37	$(\frac{2}{3}, 0, \frac{1}{3})$
38	$(\frac{2}{3}, 0, \frac{2}{3})$
39	$(\frac{1}{3}, 0, \frac{2}{3})$
40	$(0, \frac{1}{3}, \frac{1}{3})$
41	$(0, \frac{1}{3}, \frac{2}{3})$
42	$(0, \frac{2}{3}, \frac{2}{3})$
43	$(0, \frac{2}{3}, \frac{1}{3})$
44	$(1, \frac{1}{3}, \frac{1}{3})$
45	$(1, \frac{2}{3}, \frac{1}{3})$
46	$(1, \frac{2}{3}, \frac{2}{3})$
47	$(1, \frac{1}{3}, \frac{2}{3})$
48	$(\frac{1}{3}, 1, \frac{1}{3})$
49	$(\frac{1}{3}, 1, \frac{2}{3})$
50	$(\frac{2}{3}, 1, \frac{2}{3})$
51	$(\frac{2}{3}, 1, \frac{1}{3})$
52	$(\frac{1}{3}, \frac{1}{3}, 1)$
53	$(\frac{2}{3}, \frac{1}{3}, 1)$
54	$(\frac{2}{3}, \frac{2}{3}, 1)$
55	$(\frac{1}{3}, \frac{2}{3}, 1)$
56	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
57	$(\frac{2}{3}, \frac{1}{3}, \frac{1}{3})$
58	$(\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$
59	$(\frac{1}{3}, \frac{2}{3}, \frac{1}{3})$
60	$(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$
61	$(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$
62	$(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$
63	$(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$

9 Legendre polynomials

$$\begin{aligned}
P_0(\zeta) &= 1 \\
P_1(\zeta) &= \zeta \\
P_2(\zeta) &= \frac{3\zeta^2}{2} - \frac{1}{2} \\
P_3(\zeta) &= \frac{5\zeta^3}{2} - \frac{3\zeta}{2} \\
P_4(\zeta) &= \frac{35\zeta^4}{8} - \frac{15\zeta^2}{4} + \frac{3}{8} \\
P_5(\zeta) &= \frac{63\zeta^5}{8} - \frac{35\zeta^3}{4} + \frac{15\zeta}{8} \\
P_6(\zeta) &= \frac{231\zeta^6}{16} - \frac{315\zeta^4}{16} + \frac{105\zeta^2}{16} - \frac{5}{16}
\end{aligned}$$

10 Jacobi polynomials

$$\begin{aligned}
P_0^{(\alpha, \beta)}(\zeta) &= 1 \\
P_1^{(\alpha, \beta)}(\zeta) &= \frac{\alpha\zeta}{2} + \frac{\alpha}{2} + \frac{\beta\zeta}{2} - \frac{\beta}{2} + \zeta \\
P_2^{(\alpha, \beta)}(\zeta) &= -\frac{(\alpha+1)(\beta+1)(\alpha+\beta+4)}{2(\alpha+\beta+2)^2} \\
&\quad + \frac{(\alpha+\beta+3)[\alpha^2 - \beta^2 + \zeta(\alpha+\beta+2)(\alpha+\beta+4)][\alpha - \beta + (\alpha+\beta+2)\zeta]}{8(\alpha+\beta+2)^2}
\end{aligned}$$

11 Orthogonal polynomials on the reference triangle

$$\begin{aligned}
\Phi_{0,0}(\zeta, \eta) &= 1 \\
\Phi_{0,1}(\zeta, \eta) &= 3\eta - 1 \\
\Phi_{0,2}(\zeta, \eta) &= 10\eta^2 - 8\eta + 1 \\
\Phi_{0,3}(\zeta, \eta) &= 35\eta^3 - 45\eta^2 + 15\eta - 1 \\
\Phi_{1,0}(\zeta, \eta) &= \eta + 2\zeta - 1 \\
\Phi_{1,1}(\zeta, \eta) &= 5\eta^2 + 10\eta\zeta - 6\eta - 2\zeta + 1 \\
\Phi_{1,2}(\zeta, \eta) &= 21\eta^3 + 42\eta^2\zeta - 33\eta^2 - 24\eta\zeta + 13\eta + 2\zeta - 1 \\
\Phi_{2,0}(\zeta, \eta) &= \eta^2 + 6\eta\zeta - 2\eta + 6\zeta^2 - 6\zeta + 1 \\
\Phi_{2,1}(\zeta, \eta) &= 7\eta^3 + 42\eta^2\zeta - 15\eta^2 + 42\eta\zeta^2 - 48\eta\zeta + 9\eta - 6\zeta^2 + 6\zeta - 1 \\
\Phi_{3,0}(\zeta, \eta) &= \eta^3 + 12\eta^2\zeta - 3\eta^2 + 30\eta\zeta^2 - 24\eta\zeta + 3\eta + 20\zeta^3 - 30\zeta^2 + 12\zeta - 1
\end{aligned}$$

12 Orthogonal polynomials on the reference triangle in terms of barycentric coordinates

13 Orthogonal polynomials on the reference tetrahedron

$$\begin{aligned}
\Phi_{0,0,0}(\zeta, \eta, \xi) &= 1 \\
\Phi_{0,0,1}(\zeta, \eta, \xi) &= 4\xi - 1 \\
\Phi_{0,0,2}(\zeta, \eta, \xi) &= 15\xi^2 - 10\xi + 1 \\
\Phi_{0,0,3}(\zeta, \eta, \xi) &= 56\xi^3 - 63\xi^2 + 18\xi - 1 \\
\Phi_{0,1,0}(\zeta, \eta, \xi) &= 3\eta + \xi - 1 \\
\Phi_{0,1,1}(\zeta, \eta, \xi) &= 18\eta\xi - 3\eta + 6\xi^2 - 7\xi + 1 \\
\Phi_{0,1,2}(\zeta, \eta, \xi) &= 84\eta\xi^2 - 42\eta\xi + 3\eta + 28\xi^3 - 42\xi^2 + 15\xi - 1 \\
\Phi_{0,2,0}(\zeta, \eta, \xi) &= 10\eta^2 + 8\eta\xi - 8\eta + \xi^2 - 2\xi + 1 \\
\Phi_{0,2,1}(\zeta, \eta, \xi) &= 80\eta^2\xi - 10\eta^2 + 64\eta\xi^2 - 72\eta\xi + 8\eta + 8\xi^3 - 17\xi^2 + 10\xi - 1 \\
\Phi_{0,3,0}(\zeta, \eta, \xi) &= 35\eta^3 + 45\eta^2\xi - 45\eta^2 + 15\eta\xi^2 - 30\eta\xi + 15\eta + \xi^3 - 3\xi^2 + 3\xi - 1 \\
\Phi_{1,0,0}(\zeta, \eta, \xi) &= \eta + \xi + 2\zeta - 1 \\
\Phi_{1,0,1}(\zeta, \eta, \xi) &= 6\eta\xi - \eta + 6\xi^2 + 12\xi\zeta - 7\xi - 2\zeta + 1 \\
\Phi_{1,0,2}(\zeta, \eta, \xi) &= 28\eta\xi^2 - 14\eta\xi + \eta + 28\xi^3 + 56\xi^2\zeta - 42\xi^2 - 28\xi\zeta + 15\xi + 2\zeta - 1 \\
\Phi_{1,1,0}(\zeta, \eta, \xi) &= 5\eta^2 + 6\eta\xi + 10\eta\zeta - 6\eta + \xi^2 + 2\xi\zeta - 2\xi - 2\zeta + 1 \\
\Phi_{1,1,1}(\zeta, \eta, \xi) &= 40\eta^2\xi - 5\eta^2 + 48\eta\xi^2 + 80\eta\xi\zeta - 54\eta\xi - 10\eta\zeta + 6\eta + 8\xi^3 + 16\xi^2\zeta - 17\xi^2 \\
&\quad - 18\xi\zeta + 10\xi + 2\zeta - 1 \\
\Phi_{1,2,0}(\zeta, \eta, \xi) &= 21\eta^3 + 33\eta^2\xi + 42\eta^2\zeta - 33\eta^2 + 13\eta\xi^2 + 24\eta\xi\zeta - 26\eta\xi - 24\eta\zeta + 13\eta + \xi^3 \\
&\quad + 2\xi^2\zeta - 3\xi^2 - 4\xi\zeta + 3\xi + 2\zeta - 1 \\
\Phi_{2,0,0}(\zeta, \eta, \xi) &= \eta^2 + 2\eta\xi + 6\eta\zeta - 2\eta + \xi^2 + 6\xi\zeta - 2\xi + 6\zeta^2 - 6\zeta + 1 \\
\Phi_{2,0,1}(\zeta, \eta, \xi) &= 8\eta^2\xi - \eta^2 + 16\eta\xi^2 + 48\eta\xi\zeta - 18\eta\xi - 6\eta\zeta + 2\eta + 8\xi^3 + 48\xi^2\zeta - 17\xi^2 + 48\xi\zeta^2 \\
&\quad - 54\xi\zeta + 10\xi - 6\zeta^2 + 6\zeta - 1 \\
\Phi_{2,1,0}(\zeta, \eta, \xi) &= 7\eta^3 + 15\eta^2\xi + 42\eta^2\zeta - 15\eta^2 + 9\eta\xi^2 + 48\eta\xi\zeta - 18\eta\xi + 42\eta\zeta^2 - 48\eta\zeta + 9\eta + \xi^3 \\
&\quad + 6\xi^2\zeta - 3\xi^2 + 6\xi\zeta^2 - 12\xi\zeta + 3\xi - 6\zeta^2 + 6\zeta - 1 \\
\Phi_{3,0,0}(\zeta, \eta, \xi) &= \eta^3 + 3\eta^2\xi + 12\eta^2\zeta - 3\eta^2 + 3\eta\xi^2 + 24\eta\xi\zeta - 6\eta\xi + 30\eta\zeta^2 - 24\eta\zeta + 3\eta + \xi^3 \\
&\quad + 12\xi^2\zeta - 3\xi^2 + 30\xi\zeta^2 - 24\xi\zeta + 3\xi + 20\zeta^3 - 30\zeta^2 + 12\zeta - 1
\end{aligned}$$

14 Orthogonal polynomials on the reference tetrahedron in terms of barycentric coordinates

15 Nodal basis functions and their derivatives on the reference line element

15.1 P_1 reference line element

$$\begin{aligned}
\Psi_0(\zeta) &= \frac{1}{2} - \frac{\zeta}{2} \\
\Psi_1(\zeta) &= \frac{\zeta}{2} + \frac{1}{2}
\end{aligned}$$

$$\begin{aligned}\frac{d\Psi_0}{d\zeta} &= -\frac{1}{2} \\ \frac{d\Psi_1}{d\zeta} &= \frac{1}{2}\end{aligned}$$

15.2 P₂ reference line element

$$\begin{aligned}\Psi_0(\zeta) &= \frac{\zeta^2}{2} - \frac{\zeta}{2} \\ \Psi_1(\zeta) &= 1 - \zeta^2 \\ \Psi_2(\zeta) &= \frac{\zeta^2}{2} + \frac{\zeta}{2}\end{aligned}$$

$$\begin{aligned}\frac{d\Psi_0}{d\zeta} &= \zeta - \frac{1}{2} \\ \frac{d\Psi_1}{d\zeta} &= -2\zeta \\ \frac{d\Psi_2}{d\zeta} &= \zeta + \frac{1}{2}\end{aligned}$$

15.3 P₃ reference line element

$$\begin{aligned}\Psi_0(\zeta) &= -\frac{9\zeta^3}{16} + \frac{9\zeta^2}{16} + \frac{\zeta}{16} - \frac{1}{16} \\ \Psi_1(\zeta) &= \frac{27\zeta^3}{16} - \frac{9\zeta^2}{16} - \frac{27\zeta}{16} + \frac{9}{16} \\ \Psi_2(\zeta) &= -\frac{27\zeta^3}{16} - \frac{9\zeta^2}{16} + \frac{27\zeta}{16} + \frac{9}{16} \\ \Psi_3(\zeta) &= \frac{9\zeta^3}{16} + \frac{9\zeta^2}{16} - \frac{\zeta}{16} - \frac{1}{16}\end{aligned}$$

$$\begin{aligned}\frac{d\Psi_0}{d\zeta} &= -\frac{27\zeta^2}{16} + \frac{9\zeta}{8} + \frac{1}{16} \\ \frac{d\Psi_1}{d\zeta} &= \frac{81\zeta^2}{16} - \frac{9\zeta}{8} - \frac{27}{16} \\ \frac{d\Psi_2}{d\zeta} &= -\frac{81\zeta^2}{16} - \frac{9\zeta}{8} + \frac{27}{16} \\ \frac{d\Psi_3}{d\zeta} &= \frac{27\zeta^2}{16} + \frac{9\zeta}{8} - \frac{1}{16}\end{aligned}$$

16 Nodal basis functions and their derivatives on the reference triangle

16.1 P₁ reference triangle

$$\begin{aligned}\Psi_0(\zeta, \eta) &= 1 - \eta - \zeta \\ \Psi_1(\zeta, \eta) &= \zeta \\ \Psi_2(\zeta, \eta) &= \eta\end{aligned}$$

$$\frac{\partial \Psi_0}{\partial \zeta} = -1$$

$$\frac{\partial \Psi_1}{\partial \zeta} = 1$$

$$\frac{\partial \Psi_2}{\partial \zeta} = 0$$

$$\frac{\partial \Psi_0}{\partial \eta} = -1$$

$$\frac{\partial \Psi_1}{\partial \eta} = 0$$

$$\frac{\partial \Psi_2}{\partial \eta} = 1$$

16.2 P₂ reference triangle

$$\Psi_0(\zeta, \eta) = 2\eta^2 + 4\eta\zeta - 3\eta + 2\zeta^2 - 3\zeta + 1$$

$$\Psi_1(\zeta, \eta) = 2\zeta^2 - \zeta$$

$$\Psi_2(\zeta, \eta) = 2\eta^2 - \eta$$

$$\Psi_3(\zeta, \eta) = -4\eta\zeta - 4\zeta^2 + 4\zeta$$

$$\Psi_4(\zeta, \eta) = 4\eta\zeta$$

$$\Psi_5(\zeta, \eta) = -4\eta^2 - 4\eta\zeta + 4\eta$$

$$\frac{\partial \Psi_0}{\partial \zeta} = 4\eta + 4\zeta - 3$$

$$\frac{\partial \Psi_1}{\partial \zeta} = 4\zeta - 1$$

$$\frac{\partial \Psi_2}{\partial \zeta} = 0$$

$$\frac{\partial \Psi_3}{\partial \zeta} = -4\eta - 8\zeta + 4$$

$$\frac{\partial \Psi_4}{\partial \zeta} = 4\eta$$

$$\frac{\partial \Psi_5}{\partial \zeta} = -4\eta$$

$$\frac{\partial \Psi_0}{\partial \eta} = 4\eta + 4\zeta - 3$$

$$\frac{\partial \Psi_1}{\partial \eta} = 0$$

$$\frac{\partial \Psi_2}{\partial \eta} = 4\eta - 1$$

$$\frac{\partial \Psi_3}{\partial \eta} = -4\zeta$$

$$\frac{\partial \Psi_4}{\partial \eta} = 4\zeta$$

$$\frac{\partial \Psi_5}{\partial \eta} = -8\eta - 4\zeta + 4$$

16.3 P_3 reference triangle

$$\begin{aligned}
\Psi_0(\zeta, \eta) &= -\frac{9\eta^3}{2} - \frac{27\eta^2\zeta}{2} + 9\eta^2 - \frac{27\eta\zeta^2}{2} + 18\eta\zeta - \frac{11\eta}{2} - \frac{9\zeta^3}{2} + 9\zeta^2 - \frac{11\zeta}{2} + 1 \\
\Psi_1(\zeta, \eta) &= \frac{9\zeta^3}{2} - \frac{9\zeta^2}{2} + \zeta \\
\Psi_2(\zeta, \eta) &= \frac{9\eta^3}{2} - \frac{9\eta^2}{2} + \eta \\
\Psi_3(\zeta, \eta) &= \frac{27\eta^2\zeta}{2} + 27\eta\zeta^2 - \frac{45\eta\zeta}{2} + \frac{27\zeta^3}{2} - \frac{45\zeta^2}{2} + 9\zeta \\
\Psi_4(\zeta, \eta) &= -\frac{27\eta\zeta^2}{2} + \frac{9\eta\zeta}{2} - \frac{27\zeta^3}{2} + 18\zeta^2 - \frac{9\zeta}{2} \\
\Psi_5(\zeta, \eta) &= \frac{27\eta\zeta^2}{2} - \frac{9\eta\zeta}{2} \\
\Psi_6(\zeta, \eta) &= \frac{27\eta^2\zeta}{2} - \frac{9\eta\zeta}{2} \\
\Psi_7(\zeta, \eta) &= -\frac{27\eta^3}{2} - \frac{27\eta^2\zeta}{2} + 18\eta^2 + \frac{9\eta\zeta}{2} - \frac{9\eta}{2} \\
\Psi_8(\zeta, \eta) &= \frac{27\eta^3}{2} + 27\eta^2\zeta - \frac{45\eta^2}{2} + \frac{27\eta\zeta^2}{2} - \frac{45\eta\zeta}{2} + 9\eta \\
\Psi_9(\zeta, \eta) &= -27\eta^2\zeta - 27\eta\zeta^2 + 27\eta\zeta
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \Psi_0}{\partial \zeta} &= -\frac{27\eta^2}{2} - 27\eta\zeta + 18\eta - \frac{27\zeta^2}{2} + 18\zeta - \frac{11}{2} \\
\frac{\partial \Psi_1}{\partial \zeta} &= \frac{27\zeta^2}{2} - 9\zeta + 1 \\
\frac{\partial \Psi_2}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_3}{\partial \zeta} &= \frac{27\eta^2}{2} + 54\eta\zeta - \frac{45\eta}{2} + \frac{81\zeta^2}{2} - 45\zeta + 9 \\
\frac{\partial \Psi_4}{\partial \zeta} &= -27\eta\zeta + \frac{9\eta}{2} - \frac{81\zeta^2}{2} + 36\zeta - \frac{9}{2} \\
\frac{\partial \Psi_5}{\partial \zeta} &= 27\eta\zeta - \frac{9\eta}{2} \\
\frac{\partial \Psi_6}{\partial \zeta} &= \frac{27\eta^2}{2} - \frac{9\eta}{2} \\
\frac{\partial \Psi_7}{\partial \zeta} &= -\frac{27\eta^2}{2} + \frac{9\eta}{2} \\
\frac{\partial \Psi_8}{\partial \zeta} &= 27\eta^2 + 27\eta\zeta - \frac{45\eta}{2} \\
\frac{\partial \Psi_9}{\partial \zeta} &= -27\eta^2 - 54\eta\zeta + 27\eta
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \Psi_0}{\partial \eta} &= -\frac{27\eta^2}{2} - 27\eta\zeta + 18\eta - \frac{27\zeta^2}{2} + 18\zeta - \frac{11}{2} \\
\frac{\partial \Psi_1}{\partial \eta} &= 0 \\
\frac{\partial \Psi_2}{\partial \eta} &= \frac{27\eta^2}{2} - 9\eta + 1 \\
\frac{\partial \Psi_3}{\partial \eta} &= 27\eta\zeta + 27\zeta^2 - \frac{45\zeta}{2} \\
\frac{\partial \Psi_4}{\partial \eta} &= -\frac{27\zeta^2}{2} + \frac{9\zeta}{2} \\
\frac{\partial \Psi_5}{\partial \eta} &= \frac{27\zeta^2}{2} - \frac{9\zeta}{2} \\
\frac{\partial \Psi_6}{\partial \eta} &= 27\eta\zeta - \frac{9\zeta}{2} \\
\frac{\partial \Psi_7}{\partial \eta} &= -\frac{81\eta^2}{2} - 27\eta\zeta + 36\eta + \frac{9\zeta}{2} - \frac{9}{2} \\
\frac{\partial \Psi_8}{\partial \eta} &= \frac{81\eta^2}{2} + 54\eta\zeta - 45\eta + \frac{27\zeta^2}{2} - \frac{45\zeta}{2} + 9 \\
\frac{\partial \Psi_9}{\partial \eta} &= -54\eta\zeta - 27\zeta^2 + 27\zeta
\end{aligned}$$

17 Nodal basis functions and their derivatives on the reference triangle in terms of barycentric coordinates

17.1 P_1 reference triangle

17.2 P_2 reference triangle

17.3 P_3 reference triangle

18 Nodal basis functions and their derivatives on the reference tetrahedron

18.1 P_1 reference tetrahedron

$$\begin{aligned}
\Psi_0(\zeta, \eta, \xi) &= 1 - \eta - \xi - \zeta \\
\Psi_1(\zeta, \eta, \xi) &= \zeta \\
\Psi_2(\zeta, \eta, \xi) &= \eta \\
\Psi_3(\zeta, \eta, \xi) &= \xi
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \Psi_0}{\partial \zeta} &= -1 \\
\frac{\partial \Psi_1}{\partial \zeta} &= 1 \\
\frac{\partial \Psi_2}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_3}{\partial \zeta} &= 0
\end{aligned}$$

$$\frac{\partial \Psi_0}{\partial \eta} = -1$$

$$\frac{\partial \Psi_1}{\partial \eta} = 0$$

$$\frac{\partial \Psi_2}{\partial \eta} = 1$$

$$\frac{\partial \Psi_3}{\partial \eta} = 0$$

$$\frac{\partial \Psi_0}{\partial \xi} = -1$$

$$\frac{\partial \Psi_1}{\partial \xi} = 0$$

$$\frac{\partial \Psi_2}{\partial \xi} = 0$$

$$\frac{\partial \Psi_3}{\partial \xi} = 1$$

18.2 P₂ reference tetrahedron

$$\Psi_0(\zeta, \eta, \xi) = 2\eta^2 + 4\eta\xi + 4\eta\zeta - 3\eta + 2\xi^2 + 4\xi\zeta - 3\xi + 2\zeta^2 - 3\zeta + 1$$

$$\Psi_1(\zeta, \eta, \xi) = 2\zeta^2 - \zeta$$

$$\Psi_2(\zeta, \eta, \xi) = 2\eta^2 - \eta$$

$$\Psi_3(\zeta, \eta, \xi) = 2\xi^2 - \xi$$

$$\Psi_4(\zeta, \eta, \xi) = -4\eta\zeta - 4\xi\zeta - 4\zeta^2 + 4\zeta$$

$$\Psi_5(\zeta, \eta, \xi) = 4\eta\zeta$$

$$\Psi_6(\zeta, \eta, \xi) = -4\eta^2 - 4\eta\xi - 4\eta\zeta + 4\eta$$

$$\Psi_7(\zeta, \eta, \xi) = -4\eta\xi - 4\xi^2 - 4\xi\zeta + 4\xi$$

$$\Psi_8(\zeta, \eta, \xi) = 4\eta\xi$$

$$\Psi_9(\zeta, \eta, \xi) = 4\xi\zeta$$

$$\begin{aligned}
\frac{\partial \Psi_0}{\partial \zeta} &= 4\eta + 4\xi + 4\zeta - 3 \\
\frac{\partial \Psi_1}{\partial \zeta} &= 4\zeta - 1 \\
\frac{\partial \Psi_2}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_3}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_4}{\partial \zeta} &= -4\eta - 4\xi - 8\zeta + 4 \\
\frac{\partial \Psi_5}{\partial \zeta} &= 4\eta \\
\frac{\partial \Psi_6}{\partial \zeta} &= -4\eta \\
\frac{\partial \Psi_7}{\partial \zeta} &= -4\xi \\
\frac{\partial \Psi_8}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_9}{\partial \zeta} &= 4\xi
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \Psi_0}{\partial \eta} &= 4\eta + 4\xi + 4\zeta - 3 \\
\frac{\partial \Psi_1}{\partial \eta} &= 0 \\
\frac{\partial \Psi_2}{\partial \eta} &= 4\eta - 1 \\
\frac{\partial \Psi_3}{\partial \eta} &= 0 \\
\frac{\partial \Psi_4}{\partial \eta} &= -4\zeta \\
\frac{\partial \Psi_5}{\partial \eta} &= 4\zeta \\
\frac{\partial \Psi_6}{\partial \eta} &= -8\eta - 4\xi - 4\zeta + 4 \\
\frac{\partial \Psi_7}{\partial \eta} &= -4\xi \\
\frac{\partial \Psi_8}{\partial \eta} &= 4\xi \\
\frac{\partial \Psi_9}{\partial \eta} &= 0
\end{aligned}$$

$$\frac{\partial \Psi_0}{\partial \xi} = 4\eta + 4\xi + 4\zeta - 3$$

$$\frac{\partial \Psi_1}{\partial \xi} = 0$$

$$\frac{\partial \Psi_2}{\partial \xi} = 0$$

$$\frac{\partial \Psi_3}{\partial \xi} = 4\xi - 1$$

$$\frac{\partial \Psi_4}{\partial \xi} = -4\zeta$$

$$\frac{\partial \Psi_5}{\partial \xi} = 0$$

$$\frac{\partial \Psi_6}{\partial \xi} = -4\eta$$

$$\frac{\partial \Psi_7}{\partial \xi} = -4\eta - 8\xi - 4\zeta + 4$$

$$\frac{\partial \Psi_8}{\partial \xi} = 4\eta$$

$$\frac{\partial \Psi_9}{\partial \xi} = 4\zeta$$

18.3 P_3 reference tetrahedron

$$\begin{aligned}
\Psi_0(\zeta, \eta, \xi) &= -\frac{9\eta^3}{2} - \frac{27\eta^2\xi}{2} - \frac{27\eta^2\zeta}{2} + 9\eta^2 - \frac{27\eta\xi^2}{2} - 27\eta\xi\zeta + 18\eta\xi - \frac{27\eta\zeta^2}{2} + 18\eta\zeta - \frac{11\eta}{2} \\
&\quad - \frac{9\xi^3}{2} - \frac{27\xi^2\zeta}{2} + 9\xi^2 - \frac{27\xi\zeta^2}{2} + 18\xi\zeta - \frac{11\xi}{2} - \frac{9\zeta^3}{2} + 9\zeta^2 - \frac{11\zeta}{2} + 1 \\
\Psi_1(\zeta, \eta, \xi) &= \frac{9\zeta^3}{2} - \frac{9\zeta^2}{2} + \zeta \\
\Psi_2(\zeta, \eta, \xi) &= \frac{9\eta^3}{2} - \frac{9\eta^2}{2} + \eta \\
\Psi_3(\zeta, \eta, \xi) &= \frac{9\xi^3}{2} - \frac{9\xi^2}{2} + \xi \\
\Psi_4(\zeta, \eta, \xi) &= \frac{27\eta^2\zeta}{2} + 27\eta\xi\zeta + 27\eta\zeta^2 - \frac{45\eta\zeta}{2} + \frac{27\xi^2\zeta}{2} + 27\xi\zeta^2 - \frac{45\xi\zeta}{2} + \frac{27\zeta^3}{2} - \frac{45\zeta^2}{2} + 9\zeta \\
\Psi_5(\zeta, \eta, \xi) &= -\frac{27\eta\zeta^2}{2} + \frac{9\eta\zeta}{2} - \frac{27\xi\zeta^2}{2} + \frac{9\xi\zeta}{2} - \frac{27\zeta^3}{2} + 18\zeta^2 - \frac{9\zeta}{2} \\
\Psi_6(\zeta, \eta, \xi) &= \frac{27\eta\zeta^2}{2} - \frac{9\eta\zeta}{2} \\
\Psi_7(\zeta, \eta, \xi) &= \frac{27\eta^2\zeta}{2} - \frac{9\eta\zeta}{2} \\
\Psi_8(\zeta, \eta, \xi) &= -\frac{27\eta^3}{2} - \frac{27\eta^2\xi}{2} - \frac{27\eta^2\zeta}{2} + 18\eta^2 + \frac{9\eta\xi}{2} + \frac{9\eta\zeta}{2} - \frac{9\eta}{2} \\
\Psi_9(\zeta, \eta, \xi) &= \frac{27\eta^3}{2} + 27\eta^2\xi + 27\eta^2\zeta - \frac{45\eta^2}{2} + \frac{27\eta\xi^2}{2} + 27\eta\xi\zeta - \frac{45\eta\xi}{2} + \frac{27\eta\zeta^2}{2} - \frac{45\eta\zeta}{2} + 9\eta \\
\Psi_{10}(\zeta, \eta, \xi) &= -\frac{27\eta\xi^2}{2} + \frac{9\eta\xi}{2} - \frac{27\xi^3}{2} - \frac{27\xi^2\zeta}{2} + 18\xi^2 + \frac{9\xi\zeta}{2} - \frac{9\xi}{2} \\
\Psi_{11}(\zeta, \eta, \xi) &= \frac{27\eta^2\xi}{2} + 27\eta\xi^2 + 27\eta\xi\zeta - \frac{45\eta\xi}{2} + \frac{27\xi^3}{2} + 27\xi^2\zeta - \frac{45\xi^2}{2} + \frac{27\xi\zeta^2}{2} - \frac{45\xi\zeta}{2} + 9\xi \\
\Psi_{12}(\zeta, \eta, \xi) &= \frac{27\eta\xi^2}{2} - \frac{9\eta\xi}{2} \\
\Psi_{13}(\zeta, \eta, \xi) &= \frac{27\eta^2\xi}{2} - \frac{9\eta\xi}{2} \\
\Psi_{14}(\zeta, \eta, \xi) &= \frac{27\xi^2\zeta}{2} - \frac{9\xi\zeta}{2} \\
\Psi_{15}(\zeta, \eta, \xi) &= \frac{27\xi\zeta^2}{2} - \frac{9\xi\zeta}{2} \\
\Psi_{16}(\zeta, \eta, \xi) &= -27\eta^2\zeta - 27\eta\xi\zeta - 27\eta\zeta^2 + 27\eta\zeta \\
\Psi_{17}(\zeta, \eta, \xi) &= -27\eta\xi\zeta - 27\xi^2\zeta - 27\xi\zeta^2 + 27\xi\zeta \\
\Psi_{18}(\zeta, \eta, \xi) &= -27\eta^2\xi - 27\eta\xi^2 - 27\eta\xi\zeta + 27\eta\xi \\
\Psi_{19}(\zeta, \eta, \xi) &= 27\eta\xi\zeta
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \Psi_0}{\partial \zeta} &= -\frac{27\eta^2}{2} - 27\eta\xi - 27\eta\zeta + 18\eta - \frac{27\xi^2}{2} - 27\xi\zeta + 18\xi - \frac{27\zeta^2}{2} + 18\zeta - \frac{11}{2} \\
\frac{\partial \Psi_1}{\partial \zeta} &= \frac{27\zeta^2}{2} - 9\zeta + 1 \\
\frac{\partial \Psi_2}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_3}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_4}{\partial \zeta} &= \frac{27\eta^2}{2} + 27\eta\xi + 54\eta\zeta - \frac{45\eta}{2} + \frac{27\xi^2}{2} + 54\xi\zeta - \frac{45\xi}{2} + \frac{81\zeta^2}{2} - 45\zeta + 9 \\
\frac{\partial \Psi_5}{\partial \zeta} &= -27\eta\zeta + \frac{9\eta}{2} - 27\xi\zeta + \frac{9\xi}{2} - \frac{81\zeta^2}{2} + 36\zeta - \frac{9}{2} \\
\frac{\partial \Psi_6}{\partial \zeta} &= 27\eta\zeta - \frac{9\eta}{2} \\
\frac{\partial \Psi_7}{\partial \zeta} &= \frac{27\eta^2}{2} - \frac{9\eta}{2} \\
\frac{\partial \Psi_8}{\partial \zeta} &= -\frac{27\eta^2}{2} + \frac{9\eta}{2} \\
\frac{\partial \Psi_9}{\partial \zeta} &= 27\eta^2 + 27\eta\xi + 27\eta\zeta - \frac{45\eta}{2} \\
\frac{\partial \Psi_{10}}{\partial \zeta} &= -\frac{27\xi^2}{2} + \frac{9\xi}{2} \\
\frac{\partial \Psi_{11}}{\partial \zeta} &= 27\eta\xi + 27\xi^2 + 27\xi\zeta - \frac{45\xi}{2} \\
\frac{\partial \Psi_{12}}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_{13}}{\partial \zeta} &= 0 \\
\frac{\partial \Psi_{14}}{\partial \zeta} &= \frac{27\xi^2}{2} - \frac{9\xi}{2} \\
\frac{\partial \Psi_{15}}{\partial \zeta} &= 27\xi\zeta - \frac{9\xi}{2} \\
\frac{\partial \Psi_{16}}{\partial \zeta} &= -27\eta^2 - 27\eta\xi - 54\eta\zeta + 27\eta \\
\frac{\partial \Psi_{17}}{\partial \zeta} &= -27\eta\xi - 27\xi^2 - 54\xi\zeta + 27\xi \\
\frac{\partial \Psi_{18}}{\partial \zeta} &= -27\eta\xi \\
\frac{\partial \Psi_{19}}{\partial \zeta} &= 27\eta\xi
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \Psi_0}{\partial \eta} &= -\frac{27\eta^2}{2} - 27\eta\xi - 27\eta\zeta + 18\eta - \frac{27\xi^2}{2} - 27\xi\zeta + 18\xi - \frac{27\zeta^2}{2} + 18\zeta - \frac{11}{2} \\
\frac{\partial \Psi_1}{\partial \eta} &= 0 \\
\frac{\partial \Psi_2}{\partial \eta} &= \frac{27\eta^2}{2} - 9\eta + 1 \\
\frac{\partial \Psi_3}{\partial \eta} &= 0 \\
\frac{\partial \Psi_4}{\partial \eta} &= 27\eta\zeta + 27\xi\zeta + 27\zeta^2 - \frac{45\zeta}{2} \\
\frac{\partial \Psi_5}{\partial \eta} &= -\frac{27\zeta^2}{2} + \frac{9\zeta}{2} \\
\frac{\partial \Psi_6}{\partial \eta} &= \frac{27\zeta^2}{2} - \frac{9\zeta}{2} \\
\frac{\partial \Psi_7}{\partial \eta} &= 27\eta\zeta - \frac{9\zeta}{2} \\
\frac{\partial \Psi_8}{\partial \eta} &= -\frac{81\eta^2}{2} - 27\eta\xi - 27\eta\zeta + 36\eta + \frac{9\xi}{2} + \frac{9\zeta}{2} - \frac{9}{2} \\
\frac{\partial \Psi_9}{\partial \eta} &= \frac{81\eta^2}{2} + 54\eta\xi + 54\eta\zeta - 45\eta + \frac{27\xi^2}{2} + 27\xi\zeta - \frac{45\xi}{2} + \frac{27\zeta^2}{2} - \frac{45\zeta}{2} + 9 \\
\frac{\partial \Psi_{10}}{\partial \eta} &= -\frac{27\xi^2}{2} + \frac{9\xi}{2} \\
\frac{\partial \Psi_{11}}{\partial \eta} &= 27\eta\xi + 27\xi^2 + 27\xi\zeta - \frac{45\xi}{2} \\
\frac{\partial \Psi_{12}}{\partial \eta} &= \frac{27\xi^2}{2} - \frac{9\xi}{2} \\
\frac{\partial \Psi_{13}}{\partial \eta} &= 27\eta\xi - \frac{9\xi}{2} \\
\frac{\partial \Psi_{14}}{\partial \eta} &= 0 \\
\frac{\partial \Psi_{15}}{\partial \eta} &= 0 \\
\frac{\partial \Psi_{16}}{\partial \eta} &= -54\eta\zeta - 27\xi\zeta - 27\zeta^2 + 27\zeta \\
\frac{\partial \Psi_{17}}{\partial \eta} &= -27\xi\zeta \\
\frac{\partial \Psi_{18}}{\partial \eta} &= -54\eta\xi - 27\xi^2 - 27\xi\zeta + 27\xi \\
\frac{\partial \Psi_{19}}{\partial \eta} &= 27\xi\zeta
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \Psi_0}{\partial \xi} &= -\frac{27\eta^2}{2} - 27\eta\xi - 27\eta\zeta + 18\eta - \frac{27\xi^2}{2} - 27\xi\zeta + 18\xi - \frac{27\zeta^2}{2} + 18\zeta - \frac{11}{2} \\
\frac{\partial \Psi_1}{\partial \xi} &= 0 \\
\frac{\partial \Psi_2}{\partial \xi} &= 0 \\
\frac{\partial \Psi_3}{\partial \xi} &= \frac{27\xi^2}{2} - 9\xi + 1 \\
\frac{\partial \Psi_4}{\partial \xi} &= 27\eta\zeta + 27\xi\zeta + 27\zeta^2 - \frac{45\zeta}{2} \\
\frac{\partial \Psi_5}{\partial \xi} &= -\frac{27\zeta^2}{2} + \frac{9\zeta}{2} \\
\frac{\partial \Psi_6}{\partial \xi} &= 0 \\
\frac{\partial \Psi_7}{\partial \xi} &= 0 \\
\frac{\partial \Psi_8}{\partial \xi} &= -\frac{27\eta^2}{2} + \frac{9\eta}{2} \\
\frac{\partial \Psi_9}{\partial \xi} &= 27\eta^2 + 27\eta\xi + 27\eta\zeta - \frac{45\eta}{2} \\
\frac{\partial \Psi_{10}}{\partial \xi} &= -27\eta\xi + \frac{9\eta}{2} - \frac{81\xi^2}{2} - 27\xi\zeta + 36\xi + \frac{9\zeta}{2} - \frac{9}{2} \\
\frac{\partial \Psi_{11}}{\partial \xi} &= \frac{27\eta^2}{2} + 54\eta\xi + 27\eta\zeta - \frac{45\eta}{2} + \frac{81\xi^2}{2} + 54\xi\zeta - 45\xi + \frac{27\zeta^2}{2} - \frac{45\zeta}{2} + 9 \\
\frac{\partial \Psi_{12}}{\partial \xi} &= 27\eta\xi - \frac{9\eta}{2} \\
\frac{\partial \Psi_{13}}{\partial \xi} &= \frac{27\eta^2}{2} - \frac{9\eta}{2} \\
\frac{\partial \Psi_{14}}{\partial \xi} &= 27\xi\zeta - \frac{9\zeta}{2} \\
\frac{\partial \Psi_{15}}{\partial \xi} &= \frac{27\zeta^2}{2} - \frac{9\zeta}{2} \\
\frac{\partial \Psi_{16}}{\partial \xi} &= -27\eta\zeta \\
\frac{\partial \Psi_{17}}{\partial \xi} &= -27\eta\zeta - 54\xi\zeta - 27\zeta^2 + 27\zeta \\
\frac{\partial \Psi_{18}}{\partial \xi} &= -27\eta^2 - 54\eta\xi - 27\eta\zeta + 27\eta \\
\frac{\partial \Psi_{19}}{\partial \xi} &= 27\eta\zeta
\end{aligned}$$

19 Nodal basis functions and their derivatives on the reference tetrahedron in terms of barycentric coordinates

19.1 P_1 reference tetrahedron

19.2 P_2 reference tetrahedron

19.3 P_3 reference tetrahedron

20 Mesh data structures used in the FEM codes

The element numbering conventions documented in the previous sections drive a single mesh data layout that the FEM codes share across P1/P2/P3 triangles, quadrilaterals, tetrahedra and hexahedra. This section documents that layout and the conventions that wrap around it.

20.1 The five mesh arrays

Let

- N_v be the number of vertices (nodes);
- N_e be the number of volume elements (triangles or quadrilaterals in 2D; tetrahedra or hexahedra in 3D);
- N_b be the number of boundary elements on which Neumann (traction) boundary conditions are specified;
- $d \in \{2, 3\}$ be the spatial dimension;
- p_e be the number of nodes per volume element;
- p_b be the number of nodes per boundary element.

The five mesh arrays are then:

Array	Shape	Meaning
Vertices	$N_v \times d$ double	Vertices[i] holds the coordinates of the i -th node.
Elements	$N_e \times p_e$ int	Elements[i] holds the p_e global node indices of the i -th volume element, stored in <i>Gmsh native order</i> (see Sections 3–8).
BoundaryElements	$N_b \times p_b$ int	BoundaryElements[i] holds the p_b global node indices of the i -th Neumann boundary element, also in Gmsh native order.
VertexTags	N_v int	VertexTags[i] is the Dirichlet physical tag associated with the i -th node, or -1 if the node is not on any Dirichlet boundary. When a node sits on multiple Dirichlet tags, priority resolution (§20.3) selects a single winning tag.
BoundaryTags	N_b int	BoundaryTags[i] is the Neumann physical tag associated with the i -th Neumann boundary element.

Table 18: The five mesh arrays exposed by `read_gmsh_mesh()`.

All five arrays are allocated by `read_gmsh_mesh()` with `allocate()` from `general.h` (contiguous storage, raw pointers) and released by the caller with `deallocate()`; no STL container is used. Node indices are stored 0-based even though the underlying Gmsh MSH 2.2 file uses 1-based indexing.

20.2 (d, p_e, p_b) per element family

Element family	d	p_e	p_b	Gmsh volume element type	Gmsh boundary element type
P1 triangle	2	3	2	2	1
P2 triangle	2	6	3	9	8
P3 triangle	2	10	4	21	26
Q1 quadrilateral	2	4	2	3	1
Q2 quadrilateral	2	9	3	10	8
Q3 quadrilateral	2	16	4	36	26
P1 tetrahedron	3	4	3	4	2
P2 tetrahedron	3	10	6	11	9
P3 tetrahedron	3	20	10	29	21
Q1 hexahedron	3	8	4	5	3
Q2 hexahedron	3	27	9	12	10
Q3 hexahedron	3	64	16	92	36

Table 19: Spatial dimension, nodes-per-element, nodes-per-boundary-element, and the corresponding Gmsh MSH 2.2 element-type numbers.

20.3 Splitting Dirichlet and Neumann boundary information

Dirichlet information is carried at the node level (`VertexTags`), while *Neumann* information is carried at the element level (`BoundaryElements` and `BoundaryTags`). The mesh reader is told which physical tags to treat as Dirichlet and which as Neumann:

```
read_gmsh_mesh ( path,
                  d, pe, pb,
                  dirichlet_tags, n_dirichlet_tags,
                  neumann_tags,   n_neumann_tags,
                  Nv, Ne, Nb,
                  Vertices, Elements, BoundaryElements,
                  VertexTags, BoundaryTags );
```

The reader walks each boundary entity in the MSH file. If its physical tag appears in `neumann_tags`, the entity is appended to `BoundaryElements` and its tag is recorded in `BoundaryTags`. If instead its tag appears in `dirichlet_tags`, the reader updates `VertexTags` for every node of that entity. Entities whose tag appears in neither list are silently skipped (this is how traction-free boundaries remain implicit).

Priority among Dirichlet tags. When a node belongs to multiple Dirichlet boundaries, the reader resolves the conflict by listing order: earlier entries in `dirichlet_tags` take priority over later entries. The intended usage is to list the tag whose component mask is the largest (most restrictive) first. In the 3D cantilever codes, for instance, the priority order is

```
const int dirichlet_tags[] = { PHYS_DIRICHLET_FULL, /* 13 */
                              PHYS_DIRICHLET_W };   /* 14 */
```

so that a node lying on both the left clamped face and a $z = \pm d$ face inherits the fully-clamped tag rather than the w -only tag.

20.4 Per-tag physics: component masks and value functions

A single Dirichlet tag generally does not fix every displacement component (e.g. a symmetry plane fixes only the normal component). The codes therefore attach two pieces of information to each Dirichlet tag, supplied by the problem-specific source file rather than by the mesh layer:

- a *component mask* `dirichlet_mask(tag)` returning a bitmask in $\{0, \dots, 2^d - 1\}$: bit c is 1 iff the c -th DOF of nodes with this tag is prescribed;
- a *value function* `dirichlet_value(tag, x, y[, z], c)` returning the prescribed value of component c at the physical point $(x, y[, z])$.

Neumann tags carry an analogous traction function:

```
neumann_traction ( tag, x, y[, z], nx, ny[, nz],
                  tx, ty[, tz] );
```

The outward unit normal $(n_x, n_y[, n_z])$ is passed in even when the cantilever's parabolic-shear traction ignores it; this keeps the signature ready for problems where the traction is normal-dependent (pressure loads, anisotropic surface forces, etc.).

The DOF-level helpers used by the assembly routine become very short:

```
bool is_dirichlet ( int dof_idx )
{
    int n    = dof_idx / d;
    int c    = dof_idx - d*n;
    int tag = VertexTags[n];
    if (tag == -1) return false;
    return ((dirichlet_mask(tag) >> c) & 1u) != 0u;
}

double dirichlet_value_dof ( int dof_idx )
{
    int n    = dof_idx / d;
    int c    = dof_idx - d*n;
    int tag = VertexTags[n];
    return dirichlet_value(tag, Vertices[n][0], Vertices[n][1],
                          /* z if d == 3, */ c);
}
```

20.5 Gmsh native node ordering inside Elements

`Elements` and `BoundaryElements` store node indices in *Gmsh MSH 2.2 native order* verbatim. No remap is performed during reading. The implicit contract is that any kernel that reads `Elements[e][a]` must interpret index a according to the local numbering documented in Sections 3–8 of this document. The shared header `shape_functions.h` provides one Gmsh-ordered evaluator per element family:

```
shape_line2, shape_line3, shape_line4,
shape_tri3,  shape_tri6,  shape_tri10,
shape_quad4, shape_quad9, shape_quad16,
shape_tet4,  shape_tet10, shape_tet20,
shape_hex8,  shape_hex27, shape_hex64
```

Each evaluator takes a reference-space coordinate ξ and writes both the shape function values $N_a(\xi)$ and the gradients $\partial N_a / \partial \xi_j$ in Gmsh order. Tensor-product orderings used for Quad9, Quad16, Hex27 and Hex64 were verified against meshes generated by Gmsh 4.15 before being baked into the header.

20.6 Isoparametric element kernel

Each volume kernel follows the same skeleton. For each Gauss point $\xi^{(q)}$ in the reference domain:

1. Evaluate $N_a(\xi^{(q)})$ and $\partial N_a / \partial \xi_j$ via the appropriate `shape_*` routine.
2. Build the Jacobian

$$J_{ij} = \sum_{a=0}^{p_e-1} X_i^{(a)} \frac{\partial N_a}{\partial \xi_j}, \quad X^{(a)} = \text{Vertices}[\text{Elements}[e][a]].$$

3. Compute $|\det J|$ and J^{-1} .
4. Transform gradients to physical space: $\frac{\partial N_a}{\partial x_i} = \sum_j (J^{-1})_{ji} \frac{\partial N_a}{\partial \xi_j}$.
5. Build the strain-displacement matrix B (3 rows in 2D, 6 in 3D) and accumulate $w^{(q)} |\det J| B^T C B$ into the element stiffness, scattering directly into the global HYPRE matrix.

The same procedure handles affine and non-affine elements without code changes, so the v2 codes are not restricted to axis-aligned quads or hexes: the Jacobian is rebuilt at every Gauss point from `Vertices[Elements[e][a]]`. All quadrature rules are obtained from `quadrature.h` (Gauss–Legendre 1D, Dunavant-style triangle, Keast/Witherden–Vincent tetrahedron), and dynamic storage uses `allocate()` / `deallocate()` from `general.h`.

20.7 Boundary integration

The Neumann kernel walks `BoundaryElements` and uses the analogous isoparametric construction on $(d - 1)$ -dimensional surfaces.

- In 2D, each boundary element is a `Line2/Line3/Line4`. The tangent vector $d\mathbf{x}/d\xi = \sum_a (d\psi_a/d\xi) X^{(a)}$ is computed per Gauss point, and the Jacobian magnitude $|J| = \|d\mathbf{x}/d\xi\|$ is used as the arc-length element.
- In 3D, each boundary element is a `Tri3/Tri6/Tri10` or `Quad4/Quad9/Quad16`. Two tangents $\partial\mathbf{x}/\partial\xi_1$ and $\partial\mathbf{x}/\partial\xi_2$ are computed; the surface element is the magnitude of their cross product, and the outward normal (n_x, n_y, n_z) is the unit cross product (orientation up to a sign that depends on the boundary-element node ordering).

For each Gauss point the kernel evaluates the per-tag traction function and accumulates $w^{(q)} |J| \psi_a \mathbf{t}$ into the right-hand side, again scattered through the same DOF-folding helper used by the volume kernel.

20.8 Helper headers

The refactor introduced two shared headers in `Parallel/`, in addition to the pre-existing `general.h` and `quadrature.h`:

- `mesh_read_write.h`: defines `read_gmsh_mesh()` and `write_gmsh_solution()`, plus the small lookup helpers `gmsh_volume_type(d, pe)`, `gmsh_boundary_type(d, pb)` and `gmsh_nodes_per_type(d, pe, pb)` that connect the (d, p_e, p_b) table above to MSH 2.2 type numbers.
- `shape_functions.h`: provides one Gmsh-ordered shape function evaluator per element family in the $P1/P2/P3 \times \{\text{triangle, quadrilateral, tetrahedron, hexahedron}\}$ matrix, plus the line evaluators used for 2D boundaries.

Together with `general.h` (containers, allocation) and `quadrature.h` (Gauss rules) these headers define the entire mesh- and basis-handling surface that each `cantilever*_v2.cpp` consumes; the per-file source then only encodes problem-specific data (geometry, material, exact solution, per-tag plug-ins, output formatting).

21 MITC9 shell elements and verification benchmarks

Alongside the continuum `cantilever*_v2.cpp` solvers, the code base carries a separate *shell* capability: a nine-node MITC shell element with its own kernel header and a set of classical verification benchmarks. This section documents the shape-function library it shares with the continuum codes, the shell kernel `shell_mitc9.h`, and the four benchmark drivers.

21.1 The shape-function library `shape_functions.h`

`Parallel/shape_functions.h` provides, for every Gmsh element family used in the code base, one evaluator that returns the shape-function values and their reference-space gradients at a point. Each evaluator has the uniform signature

```
void shape_xxx ( const double xi[d], double N[pe], double dNdx[pe][d] );
```

where `d` is the reference dimension, `pe` the node count, `xi` the reference coordinates, `N` the shape-function values and `dNdx` the gradients $\partial N_a / \partial \xi_j$.

Family	Evaluators (orders 1–3)	Gmsh types
Line	<code>shape_line2/3/4</code>	1, 8, 26
Triangle	<code>shape_tri3/6/10</code>	2, 9, 21
Quadrilateral	<code>shape_quad4/9/16</code>	3, 10, 36
Tetrahedron	<code>shape_tet4/10/20</code>	4, 11, 29
Hexahedron	<code>shape_hex8/27/64</code>	5, 12, 92

The tensor-product families (quadrilateral, hexahedron) are assembled from the one-dimensional Lagrange polynomials `lagrange1d_p1/p2/p3` on $[-1, 1]$; the simplex families (triangle, tetrahedron) use barycentric coordinates $L_0 = 1 - \sum \xi_i$, $L_1 = \xi_1, \dots$. Node orderings match the Gmsh MSH 2.2 native ordering verbatim, so no remapping is needed between the mesh reader and the element kernels. The reference domains are the unit simplex for triangles and tetrahedra and $[-1, 1]^d$ for quadrilaterals and hexahedra.

21.1.1 One-dimensional Lagrange basis

Every tensor-product evaluator is built from the one-dimensional Lagrange polynomials on the reference interval $[-1, 1]$, returned together with their derivatives by `lagrange1d_p1/p2/p3`:

$$\begin{aligned}
p = 1: \quad & L_0 = \frac{1}{2}(1 - \xi), \quad L_1 = \frac{1}{2}(1 + \xi), \\
p = 2: \quad & L_0 = \frac{1}{2}\xi(\xi - 1), \quad L_1 = 1 - \xi^2, \quad L_2 = \frac{1}{2}\xi(\xi + 1), \\
p = 3: \quad & L_a = \text{cubic interpolant through } \xi = -1, -\frac{1}{3}, +\frac{1}{3}, +1.
\end{aligned}$$

Node m of order p sits at $\xi_m = -1 + 2m/p$, and the basis is interpolatory, $L_a(\xi_m) = \delta_{am}$. Each routine returns both L_a and $dL_a/d\xi$.

21.1.2 Tensor-product families (quadrilateral, hexahedron)

For a d -dimensional tensor-product element of order p , every node a carries a lattice multi-index $\alpha^a \in \{0, \dots, p\}^d$ stored in a per-family table (for example `A[9][2]` inside `shape_quad9`); that table is what encodes the Gmsh node ordering. The shape function and its reference gradient are the separable products

$$N_a(\boldsymbol{\xi}) = \prod_{m=1}^d L_{\alpha_m^a}^{(p)}(\xi_m), \quad \frac{\partial N_a}{\partial \xi_n} = L_{\alpha_n^a}^{(p)}(\xi_n) \prod_{m \neq n} L_{\alpha_m^a}^{(p)}(\xi_m). \quad (1)$$

```

1: input reference point  $\boldsymbol{\xi} = (\xi_1, \dots, \xi_d)$ 
2: for each direction  $m = 1, \dots, d$  do
3:   evaluate  $(L_m[\cdot], L'_m[\cdot]) \leftarrow \text{lagrange1d\_p}(\xi_m)$ 
4: end for
5: for each node  $a$  do
6:    $N_a \leftarrow \prod_m L_m[\alpha_m^a]$ 
7:   for each direction  $n = 1, \dots, d$  do
8:      $\partial N_a / \partial \xi_n \leftarrow L'_n[\alpha_n^a] \cdot \prod_{m \neq n} L_m[\alpha_m^a]$ 
9:   end for
10: end for
```

21.1.3 Simplex families (triangle, tetrahedron)

The simplex evaluators use barycentric coordinates $L_0 = 1 - \sum_i \xi_i$ and $L_i = \xi_i$. Shape functions are explicit polynomials in L — for the six-node triangle the vertex functions are $L_i(2L_i - 1)$ and the mid-edge functions $4L_i L_j$; higher orders follow the cubic and quartic patterns coded in `shape_tri10`, `shape_tet20`, etc. Because L_0 depends on *every* ξ_i , the reference gradient cannot be read off directly: the evaluators first compute the barycentric derivatives $\partial N_a / \partial L_i$ for all i , then apply the chain rule

$$\frac{\partial N_a}{\partial \xi_j} = \frac{\partial N_a}{\partial L_j} - \frac{\partial N_a}{\partial L_0}, \quad j = 1, \dots, d, \quad (2)$$

which accounts for $\partial L_0 / \partial \xi_j = -1$ and $\partial L_i / \partial \xi_j = \delta_{ij}$.

The MITC9 shell kernel reuses exactly one evaluator from this library: `shape_quad9` (the biquadratic nine-node quadrilateral, Gmsh type 10), which supplies the in-plane interpolation of the shell mid-surface.

21.2 The MITC9 shell kernel `shell_mitc9.h`

`Parallel/shell_mitc9.h` implements a nine-node *MITC* shell element (Mixed Interpolation of Tensorial Components, Bucelem & Bathe 1993). It is a continuum-based (degenerated-solid) shell with five degrees of freedom per node,

$$(u_x, u_y, u_z, \alpha, \beta), \quad (3)$$

the three translations of the mid-surface node plus two rotations α, β of the shell director about the nodal triad vectors $\mathbf{V}_1, \mathbf{V}_2$. The global degree of freedom of node n , component c , is $\text{dof}(n, c) = 5n + c$.

Kinematics. With $N_k(r, s)$ the biquadratic shape functions, $\mathbf{X}_k$ the nodal coordinates, a_k the nodal thickness, $\mathbf{V}_n^k$ the unit director and $t \in [-1, 1]$ the thickness coordinate, the geometry

and displacement fields are

$$\mathbf{x}(r, s, t) = \sum_k N_k(r, s) \left[\mathbf{X}_k + \frac{t a_k}{2} \mathbf{V}_n^k \right], \quad (4)$$

$$\mathbf{u}(r, s, t) = \sum_k N_k(r, s) \left[\mathbf{u}_k + \frac{t a_k}{2} (-\alpha_k \mathbf{V}_2^k + \beta_k \mathbf{V}_1^k) \right]. \quad (5)$$

The covariant base vectors are $\mathbf{g}_i = \partial \mathbf{x} / \partial \xi_i$ with $\xi_i \in \{r, s, t\}$, and the covariant strain components are

$$e_{ij} = \frac{1}{2} (\mathbf{g}_i \cdot \mathbf{u}_{,j} + \mathbf{g}_j \cdot \mathbf{u}_{,i}). \quad (6)$$

MITC assumed strains. To remove both transverse-shear and membrane locking, all five strain components are re-interpolated from tying points. The strains are first rotated into a single fixed element-centre Cartesian frame, so a constant strain tensor has constant components that any partition-of-unity tying reproduces exactly; a mean correction $\Delta = V^{-1} \int (\mathbf{B}_{\text{compatible}} - \mathbf{B}_{\text{tied}}) dV$ is then added so the assumed operator also satisfies the element-average consistency condition. Together these make the element pass the distorted-mesh patch test. The tying layout (in centre-frame Cartesian strains) is

- $\varepsilon_{11}, \gamma_{13}$: linear in r at $r = \pm 1/\sqrt{3}$, quadratic in s (2×3 points);
- $\varepsilon_{22}, \gamma_{23}$: quadratic in r , linear in s at $s = \pm 1/\sqrt{3}$ (3×2 points);
- γ_{12} : bilinear at $r, s = \pm 1/\sqrt{3}$ (2×2 points).

The constitutive law is plane-stress in the lamina frame, with a transverse shear correction factor $5/6$, and the element stiffness $\mathbf{K}_e = \int \mathbf{B}^T \mathbf{D} \mathbf{B} |\det J| dr ds dt$ is integrated with a 3×3 in-plane and 2-point through-thickness Gauss rule.

Header layout. The kernel is organised in seven sections:

§	Contents
1	Fixed-size linear algebra helpers (3-vectors, 3×3 inverse).
2	Shell model data: coordinates, connectivity, thickness, triads, material.
3	Nodal director triads $\mathbf{V}_n, \mathbf{V}_1, \mathbf{V}_2$ from the geometry.
4	Constitutive matrix and covariant strain-displacement operator.
5	MITC9 assumed-strain interpolation and the 45×45 element stiffness.
6	Load vector, boundary conditions, HYPRE assembly and solve.
7	Structured <code>quad9</code> mesh generator (open and periodic).

The global system $\mathbf{K} \mathbf{u} = \mathbf{F}$ is solved with the same HYPRE stack as the continuum codes — conjugate gradients preconditioned by BoomerAMG. Prescribed degrees of freedom are handled by static condensation into the right-hand side together with identity rows. Section 7 generates a structured grid of MITC9 elements over a logical unit square and lets each driver supply the map $(u, v) \mapsto (x, y, z)$ that places the mid-surface in $\mathbb{R}^3$; `shell_build_structured_periodic` additionally wraps the u direction so that closed surfaces of revolution carry no seam.

21.2.1 Nodal director triad construction

`shell_build_directors` computes, at every node, the unit director $\mathbf{V}_n$ and the complementary triad $\mathbf{V}_1, \mathbf{V}_2$ that the rotational degrees of freedom act about. The director is the average of the mid-surface normals of all elements meeting at the node; the triad is then completed so that $\mathbf{V}_1 \times \mathbf{V}_2 = \mathbf{V}_n$.

```

1:  $\mathbf{V}_n^n \leftarrow \mathbf{0}$  for every node  $n$ 
2: for each element  $e$  do
3:   for each local node  $a = 0, \dots, 8$  do
4:      $(r, s) \leftarrow$  reference coordinates of node  $a$ 
5:      $\mathbf{g}_r \leftarrow \sum_b N_{b,r}(r, s) \mathbf{X}_b, \quad \mathbf{g}_s \leftarrow \sum_b N_{b,s}(r, s) \mathbf{X}_b$ 
6:      $\mathbf{n} \leftarrow (\mathbf{g}_r \times \mathbf{g}_s) / \|\mathbf{g}_r \times \mathbf{g}_s\|$ 
7:      $\mathbf{V}_n^{\text{global}(e,a)} += \mathbf{n}$ 
8:   end for
9: end for
10: for each node  $n$  do
11:   normalize  $\mathbf{V}_n^n$ 
12:    $\mathbf{s} \leftarrow$  the global axis  $\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$  least aligned with  $\mathbf{V}_n^n$ 
13:    $\mathbf{V}_1^n \leftarrow (\mathbf{s} \times \mathbf{V}_n^n) / \|\cdot\|, \quad \mathbf{V}_2^n \leftarrow (\mathbf{V}_n^n \times \mathbf{V}_1^n) / \|\cdot\|$ 
14: end for
```

Choosing the seed $\mathbf{s}$ as the least-aligned global axis keeps the cross product well conditioned for any director orientation.

21.2.2 Covariant base vectors and the strain-displacement operator

At a natural point (r, s, t) the covariant base vectors of the degenerated-solid geometry are

$$\mathbf{g}_r = \sum_k N_{k,r} [\mathbf{X}_k + th_k \mathbf{V}_n^k], \quad \mathbf{g}_s = \sum_k N_{k,s} [\mathbf{X}_k + th_k \mathbf{V}_n^k], \quad \mathbf{g}_t = \sum_k N_k h_k \mathbf{V}_n^k, \quad (7)$$

with $h_k = a_k/2$ the nodal half-thickness. For node k define the rotational coefficient vectors

$$\begin{aligned} \mathbf{c}_r^A &= th_k N_{k,r} (-\mathbf{V}_2^k), & \mathbf{c}_r^B &= th_k N_{k,r} \mathbf{V}_1^k, \\ \mathbf{c}_s^A &= th_k N_{k,s} (-\mathbf{V}_2^k), & \mathbf{c}_s^B &= th_k N_{k,s} \mathbf{V}_1^k, \\ \mathbf{c}_t^A &= h_k N_k (-\mathbf{V}_2^k), & \mathbf{c}_t^B &= h_k N_k \mathbf{V}_1^k. \end{aligned}$$

The contribution of node k 's five degrees of freedom $(u_x, u_y, u_z, \alpha, \beta)$ to the tensor covariant strains $e_{ij} = \frac{1}{2}(\mathbf{g}_i \cdot \mathbf{u}_{,j} + \mathbf{g}_j \cdot \mathbf{u}_{,i})$ is then, with the translational block written as a 3-vector of (u_x, u_y, u_z) coefficients,

$$\begin{aligned} e_{rr} &: [N_{k,r} \mathbf{g}_r \mid \mathbf{g}_r \cdot \mathbf{c}_r^A \mid \mathbf{g}_r \cdot \mathbf{c}_r^B], \\ e_{ss} &: [N_{k,s} \mathbf{g}_s \mid \mathbf{g}_s \cdot \mathbf{c}_s^A \mid \mathbf{g}_s \cdot \mathbf{c}_s^B], \\ e_{rs} &: \frac{1}{2} [N_{k,s} \mathbf{g}_r + N_{k,r} \mathbf{g}_s \mid \mathbf{g}_r \cdot \mathbf{c}_s^A + \mathbf{g}_s \cdot \mathbf{c}_r^A \mid \mathbf{g}_r \cdot \mathbf{c}_s^B + \mathbf{g}_s \cdot \mathbf{c}_r^B], \\ e_{rt} &: \frac{1}{2} [N_{k,r} \mathbf{g}_t \mid \mathbf{g}_r \cdot \mathbf{c}_t^A + \mathbf{g}_t \cdot \mathbf{c}_r^A \mid \mathbf{g}_r \cdot \mathbf{c}_t^B + \mathbf{g}_t \cdot \mathbf{c}_r^B], \\ e_{st} &: \frac{1}{2} [N_{k,s} \mathbf{g}_t \mid \mathbf{g}_s \cdot \mathbf{c}_t^A + \mathbf{g}_t \cdot \mathbf{c}_s^A \mid \mathbf{g}_s \cdot \mathbf{c}_t^B + \mathbf{g}_t \cdot \mathbf{c}_s^B]. \end{aligned} \quad (8)$$

These five rows, assembled over all nine nodes, form the 5×45 displacement-based covariant operator $\mathbf{B}^{\text{cov}}$.

21.2.3 Lamina frame and the covariant-to-Cartesian transform

The constitutive law is applied in a local Cartesian *lamina* frame, so the covariant strains are rotated into that frame by a 5×5 transform $\mathbf{T}$.

- 1: $\mathbf{e}_3 \leftarrow (\mathbf{g}_r \times \mathbf{g}_s)/\|\cdot\|$ ▷ mid-surface normal
- 2: $\mathbf{e}_1 \leftarrow \mathbf{g}_r - (\mathbf{g}_r \cdot \mathbf{e}_3)\mathbf{e}_3$; normalize ▷ in-plane, along $\mathbf{g}_r$
- 3: $\mathbf{e}_2 \leftarrow \mathbf{e}_3 \times \mathbf{e}_1$
- 4: $[\mathbf{g}^r \ \mathbf{g}^s \ \mathbf{g}^t]^\top \leftarrow [\mathbf{g}_r \ \mathbf{g}_s \ \mathbf{g}_t]^{-1}$ ▷ contravariant base vectors
- 5: $q_{ai} \leftarrow \mathbf{e}_a \cdot \mathbf{g}^i$, $a \in \{1, 2, 3\}$, $i \in \{r, s, t\}$
- 6: assemble $\mathbf{T}$ row by row from $\varepsilon_{ab} = \sum_{ij} q_{ai} q_{bj} e_{ij}$ (engineering shear strains doubled, e_{tt} dropped)

The Cartesian strain-displacement operator is $\mathbf{B} = \mathbf{T} \mathbf{B}^{\text{cov}}$, ordered as $[\varepsilon_{11}, \varepsilon_{22}, \gamma_{12}, \gamma_{13}, \gamma_{23}]$, and the plane-stress shell constitutive matrix is

$$\mathbf{D} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 & 0 & 0 \\ \nu & 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1-\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & k \frac{1-\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & k \frac{1-\nu}{2} \end{bmatrix}, \quad k = \frac{5}{6}. \quad (9)$$

21.2.4 Element stiffness: assumed strains and the mean correction

The element routine builds the locking-free assumed-strain operator and forms $\mathbf{K}_e$. All tying is done in a single *fixed* Cartesian frame ($\mathbf{e}_1^c, \mathbf{e}_2^c, \mathbf{e}_3^c$) taken at the element centre, so that a constant strain tensor has constant components and is reproduced exactly by the partition-of-unity tying interpolation.

- 1: $\mathbf{D} \leftarrow$ constitutive matrix; $\mathbf{K}_e \leftarrow \mathbf{0}$
- 2: base vectors at centre $(0, 0, 0) \Rightarrow$ fixed frame ($\mathbf{e}_1^c, \mathbf{e}_2^c, \mathbf{e}_3^c$)
- 3: Gauss rules: 3×3 in-plane, 2-point through thickness (18 points total)
- 4: $\mathbf{S} \leftarrow \mathbf{0}$, $V \leftarrow 0$ ▷ running $\int (\mathbf{B}^{\text{cmp}} - \mathbf{B}^{\text{tied}})$
- 5: **for** each thickness station t_g **do**
- 6: pre-evaluate the fixed-frame Cartesian $\mathbf{B}$ at the tying points:
 $\mathbf{B}^A$ at 2×3 points $(\varepsilon_{11}, \gamma_{13})$, $\mathbf{B}^B$ at 3×2 $(\varepsilon_{22}, \gamma_{23})$, $\mathbf{B}^C$ at 2×2 (γ_{12})
- 7: **for** each in-plane Gauss point (r_g, s_g) **do**
- 8: $\mathbf{B}^{\text{cmp}} \leftarrow$ fixed-frame Cartesian $\mathbf{B}$ at (r_g, s_g, t_g)
- 9: $JW \leftarrow \det[\mathbf{g}_r \ \mathbf{g}_s \ \mathbf{g}_t] \cdot w_g$
- 10: $\mathbf{B}^{\text{tied}}$ rows $\varepsilon_{11}, \gamma_{13} \leftarrow \sum \text{lin}(r_g) \text{quad}(s_g) \mathbf{B}^A$
- 11: $\mathbf{B}^{\text{tied}}$ rows $\varepsilon_{22}, \gamma_{23} \leftarrow \sum \text{quad}(r_g) \text{lin}(s_g) \mathbf{B}^B$
- 12: $\mathbf{B}^{\text{tied}}$ row $\gamma_{12} \leftarrow \sum \text{lin}(r_g) \text{lin}(s_g) \mathbf{B}^C$
- 13: store $(\mathbf{B}^{\text{tied}}, JW)$; $\mathbf{S} += (\mathbf{B}^{\text{cmp}} - \mathbf{B}^{\text{tied}}) JW$; $V += JW$
- 14: **end for**
- 15: **end for**
- 16: mean correction $\Delta \leftarrow \mathbf{S}/V$
- 17: **for** each Gauss point g **do**
- 18: $\mathbf{B} \leftarrow \mathbf{B}_g^{\text{tied}} + \Delta$
- 19: $\mathbf{K}_e += \mathbf{B}^\top \mathbf{D} \mathbf{B} JW_g$
- 20: **end for**

The 1D weights $\text{lin}(\cdot)$ and $\text{quad}(\cdot)$ are Lagrange interpolants through the tying abscissae $\pm 1/\sqrt{3}$ and $-\sqrt{0.6}, 0, \sqrt{0.6}$. The mean correction Δ enforces the element-average consistency condition $\int_{\Omega_e} \mathbf{B} dV = \int_{\Omega_e} \mathbf{B}^{\text{cmp}} dV$: it *vanishes* for any constant strain state (there $\mathbf{B}^{\text{tied}} \equiv \mathbf{B}^{\text{cmp}}$ in the fixed frame, so $\mathbf{S} = \mathbf{0}$), hence the locking cure of the tying is preserved while the patch test is satisfied exactly.

21.2.5 Global assembly and solution

`shell_solve` assembles $\mathbf{Ku} = \mathbf{F}$ into HYPRE IJ objects and solves it. Prescribed degrees of freedom are removed by static condensation rather than by penalty.

```

1: for each element  $e$  do
2:    $\mathbf{K}_e \leftarrow \text{shell\_element\_stiffness}(e)$ ;  $\mathbf{g} \leftarrow$  global DOF map of  $e$ 
3:   for each local pair  $(p, q)$  with free row  $g_p$  do
4:     if column  $g_q$  is prescribed then
5:        $b_{g_p} -= K_{e,pq} \cdot \bar{u}_{g_q}$  ▷ move known term to RHS
6:     else
7:        $A_{g_p, g_q} += K_{e,pq}$ 
8:     end if
9:   end for
10: end for
11: add external loads  $F_d$  on every free DOF  $d$ 
12: for each prescribed DOF  $d$  do
13:    $A_{dd} \leftarrow 1$ ,  $b_d \leftarrow \bar{u}_d$  ▷ identity row
14: end for
15: solve with conjugate gradients preconditioned by BoomerAMG

```

The structured generator of §7 (`shell_build_structured` and its periodic variant) builds the connectivity in `shape_quad9` lattice order and assigns node identities $i + j(2n_x + 1)$, wrapping the i -index modulo $2n_x$ when a closed surface of revolution is requested.

21.3 The four shell verification benchmarks

Four driver programs in the `Codes/` directory exercise the element on the standard linear shell benchmarks. Each generates its own mesh internally (no external mesher is required), runs a mesh-convergence sweep, and prints the measured quantity against the published reference value. Passing an `argv` argument flips `shell::use_tying` off, replacing the assumed strains by the pure displacement field — a direct demonstration of locking.

Benchmark	Configuration	Reference
Cook's membrane <code>cooks_membrane_m</code>	Tapered panel, corners (0, 0), (48, 44), (48, 60), (0, 44); $E = 1$, $\nu = 1/3$, $t = 1$; clamped left edge; unit shear load on the right edge. Measured: tip vertical displacement.	$u_y \approx 23.96$ at the loaded-edge midpoint
Scordelis–Lo roof <code>scordelis_lo_mit</code>	Barrel vault, $R = 25$, $L = 50$, $t = 0.25$, 80° arc; $E = 4.32 \times 10^8$, $\nu = 0$; rigid end diaphragms; self-weight $q = 90$. Measured: free-edge vertical deflection.	0.3024
Pinched cylinder <code>pinched_cylinder</code>	Closed cylinder, $R = 300$, $L = 600$, $t = 3$; $E = 3 \times 10^6$, $\nu = 0.3$; rigid end diaphragms; two opposite unit radial point loads at mid-length. Measured: radial displacement under a load.	1.8248×10^{-5}
Hemispherical shell <code>hemisphere_mitc9</code>	Hemisphere with an 18° polar hole, $R = 10$, $t = 0.04$; $E = 6.825 \times 10^7$, $\nu = 0.3$; four alternating radial point loads $P = 2$ at the equator. Measured: radial displacement at a loaded point.	0.094

The Scordelis–Lo roof is modelled as the full vault (no symmetry planes are imposed) so the diaphragm boundary conditions stay unambiguous; the pinched cylinder and the hemisphere use the periodic mesh generator to model the full closed surface. In every case the residual rigid-body modes are removed by a minimal set of additional point constraints, as documented in the header of each driver. Two further drivers, `patch_test_mitc9.cpp` and `patch_test2_mitc9.cpp`, run the constant-strain patch test on regular and distorted meshes and provide a mesh-independent correctness check that complements the four convergence benchmarks above.

Each driver is built and run exactly like the continuum solvers, for example

```
mpic++ -O2 -I/home/pritam-giri/HYPRE/include scordelis_lo_mitc9.cpp \
-L/home/pritam-giri/HYPRE/lib -lHYPRE -lm -o run_scordelis
LD_LIBRARY_PATH=/home/pritam-giri/HYPRE/lib:/home/pritam-giri/MPI/lib \
./run_scordelis
```

A Converting a multiple integral on a transformed domain

When evaluating element integrals, every term in an FEM code is pulled back from a physical element Ω_e to a fixed reference element $\hat{\Omega}$ via a smooth, invertible mapping. The compensation for the stretching/squeezing of volume under that mapping is the *Jacobian determinant*.

General change-of-variables formula

Let $\mathbf{x} = \boldsymbol{\varphi}(\boldsymbol{\xi})$ be a C^1 bijection from $\hat{\Omega} \subset \mathbb{R}^n$ onto $\Omega \subset \mathbb{R}^n$ with Jacobian matrix

$$J(\boldsymbol{\xi}) = \frac{\partial \mathbf{x}}{\partial \boldsymbol{\xi}} = \left[\frac{\partial x_i}{\partial \xi_j} \right]_{i,j=1}^n. \quad (10)$$

Then for any integrable $f : \Omega \rightarrow \mathbb{R}$,

$$\boxed{\int_{\Omega} f(\mathbf{x}) \, d\mathbf{x} = \int_{\hat{\Omega}} f(\varphi(\boldsymbol{\xi})) |\det J(\boldsymbol{\xi})| \, d\boldsymbol{\xi}.} \quad (11)$$

Proof of the change-of-variables formula

The argument proceeds in three steps: first for linear maps, then by local linearization on small cells, and finally by a partition-of-unity / density argument for general integrands.

Step 1: linear maps scale volume by $|\det A|$. Let $A \in \mathbb{R}^{n \times n}$ be invertible and consider the linear map $T(\boldsymbol{\xi}) = A\boldsymbol{\xi}$. For any measurable set $\hat{E} \subset \mathbb{R}^n$,

$$\text{vol}(A(\hat{E})) = |\det A| \text{vol}(\hat{E}). \quad (12)$$

This is a classical consequence of the fact that every invertible A factors as a finite product of elementary matrices (row swaps, scalings, shears). Each elementary factor either

- swaps two coordinates (multiplies volume by $|-1| = 1$, and $|\det| = 1$);
- scales one coordinate by λ (multiplies volume by $|\lambda|$, and $|\det| = |\lambda|$); or
- adds a multiple of one coordinate to another — a shear — (preserves volume by Cavalieri's principle, and $|\det| = 1$).

Taking the product of these factors gives (12). Equivalently, for the indicator function $\chi_{A(\hat{E})}$,

$$\int_{\mathbb{R}^n} \chi_{A(\hat{E})}(\mathbf{x}) \, d\mathbf{x} = \int_{\mathbb{R}^n} \chi_{\hat{E}}(\boldsymbol{\xi}) |\det A| \, d\boldsymbol{\xi}, \quad (13)$$

which is (11) in the linear, indicator-function case.

Step 2: local linearization. Now let $\varphi : \hat{\Omega} \rightarrow \Omega$ be a C^1 diffeomorphism. Fix $\boldsymbol{\xi}_0 \in \hat{\Omega}$. By differentiability,

$$\varphi(\boldsymbol{\xi}) = \varphi(\boldsymbol{\xi}_0) + J(\boldsymbol{\xi}_0)(\boldsymbol{\xi} - \boldsymbol{\xi}_0) + \mathbf{r}(\boldsymbol{\xi}), \quad \frac{\|\mathbf{r}(\boldsymbol{\xi})\|}{\|\boldsymbol{\xi} - \boldsymbol{\xi}_0\|} \xrightarrow{\boldsymbol{\xi} \rightarrow \boldsymbol{\xi}_0} 0. \quad (14)$$

Partition $\hat{\Omega}$ into a fine grid of cubes $\{\hat{Q}_k\}$ of side h with centres $\boldsymbol{\xi}_k$. On each $\hat{Q}_k$ the map φ differs from its affine approximation $T_k(\boldsymbol{\xi}) := \varphi(\boldsymbol{\xi}_k) + J(\boldsymbol{\xi}_k)(\boldsymbol{\xi} - \boldsymbol{\xi}_k)$ by a term of size $o(h)$ uniformly in k , so the image $\varphi(\hat{Q}_k)$ has volume

$$\text{vol}(\varphi(\hat{Q}_k)) = \text{vol}(T_k(\hat{Q}_k)) + o(h^n) \stackrel{(12)}{=} |\det J(\boldsymbol{\xi}_k)| \text{vol}(\hat{Q}_k) + o(h^n). \quad (15)$$

Step 3: Riemann sums and passage to the limit. Let f be continuous on Ω (the general L^1 case follows by density). Because φ is a diffeomorphism, the images $\{\varphi(\hat{Q}_k)\}$ tile Ω , so

$$\int_{\Omega} f(\mathbf{x}) \, d\mathbf{x} = \lim_{h \rightarrow 0} \sum_k f(\varphi(\boldsymbol{\xi}_k)) \text{vol}(\varphi(\hat{Q}_k)). \quad (16)$$

Substituting (15) and noting that the $o(h^n)$ remainders sum to $o(1)$ over the $O(h^{-n})$ cells in a bounded domain,

$$\int_{\Omega} f(\mathbf{x}) \, d\mathbf{x} = \lim_{h \rightarrow 0} \sum_k f(\varphi(\boldsymbol{\xi}_k)) |\det J(\boldsymbol{\xi}_k)| \text{vol}(\hat{Q}_k). \quad (17)$$

The right-hand side is precisely the Riemann sum, on the partition $\{\hat{Q}_k\}$ of $\hat{\Omega}$, of the continuous integrand $\boldsymbol{\xi} \mapsto f(\boldsymbol{\varphi}(\boldsymbol{\xi})) |\det J(\boldsymbol{\xi})|$. Taking $h \rightarrow 0$ gives

$$\int_{\Omega} f(\mathbf{x}) \, d\mathbf{x} = \int_{\hat{\Omega}} f(\boldsymbol{\varphi}(\boldsymbol{\xi})) |\det J(\boldsymbol{\xi})| \, d\boldsymbol{\xi}, \quad (18)$$

which is (11). The extension from continuous f to general integrable f is the standard one: approximate f in L^1 by continuous functions of compact support and pass to the limit on both sides, using that $\boldsymbol{\xi} \mapsto |\det J(\boldsymbol{\xi})|$ is bounded on compact subsets of $\hat{\Omega}$. ■

Common low-dimensional cases

- **2D** ($n = 2$): with $x = x(\xi, \eta)$, $y = y(\xi, \eta)$,

$$\iint_{\Omega} f \, dx \, dy = \iint_{\hat{\Omega}} f(x(\xi, \eta), y(\xi, \eta)) |x_{\xi} y_{\eta} - x_{\eta} y_{\xi}| \, d\xi \, d\eta. \quad (19)$$

- **3D**: identical structure with the 3×3 Jacobian determinant.
- **Polar**: $dx \, dy = r \, dr \, d\theta$.
- **Spherical**: $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.

FEM (isoparametric) setting

In an isoparametric formulation, the geometry of element Ω_e is described by the same shape functions $N_a(\boldsymbol{\xi})$ as the solution field:

$$\mathbf{x}(\boldsymbol{\xi}) = \sum_a N_a(\boldsymbol{\xi}) \mathbf{x}_a, \quad J_e(\boldsymbol{\xi}) = \frac{\partial \mathbf{x}}{\partial \boldsymbol{\xi}}. \quad (20)$$

Equation (11) then specializes to

$$\int_{\Omega_e} f(\mathbf{x}) \, d\Omega = \int_{\hat{\Omega}} f(\mathbf{x}(\boldsymbol{\xi})) |\det J_e(\boldsymbol{\xi})| \, d\hat{\Omega}, \quad (21)$$

and the reference-element integral is finally approximated by a quadrature rule (Gauss, Gauss–Lobatto, etc.):

$$\int_{\hat{\Omega}} g(\boldsymbol{\xi}) \, d\hat{\Omega} \approx \sum_{q=1}^{n_q} w_q g(\boldsymbol{\xi}_q). \quad (22)$$

Transformation of derivatives

Stiffness-type integrands involve gradients with respect to the physical coordinates. These transform via the inverse Jacobian:

$$\nabla_{\mathbf{x}} N_a = J^{-T} \nabla_{\boldsymbol{\xi}} N_a, \quad (23)$$

so a representative stiffness term becomes

$$\int_{\Omega_e} (\nabla N_a) \cdot (\nabla N_b) \, d\Omega = \int_{\hat{\Omega}} (J^{-T} \nabla_{\boldsymbol{\xi}} N_a) \cdot (J^{-T} \nabla_{\boldsymbol{\xi}} N_b) |\det J| \, d\hat{\Omega}. \quad (24)$$

Requirements and pitfalls

- φ must be one-to-one on the interior and C^1 .
- $\det J \neq 0$ throughout $\hat{\Omega}$ (no element inversion or folding).
- Use the *absolute value* $|\det J|$ unless orientation is being tracked (signed integrals, differential forms).
- For surface integrals in $\mathbb{R}^3$, replace $|\det J|$ with the Gram determinant $dS = \|\mathbf{x}_\xi \times \mathbf{x}_\eta\| d\xi d\eta$.